

TOPICAL REVIEW

AI Innovations in rPPG Systems for Driver Monitoring: Comprehensive Systematic Review and Future Prospects

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ABSTRACT Advanced technologies, notably camera-based systems using remote photoplethysmography (rPPG), are increasingly used in automotive safety to non-invasively monitor driver well-being and fatigue by measuring physiological metrics like heart and respiration rates. This review examines recent advancements in machine learning algorithms and signal processing for rPPG in driver monitoring. A literature search up to April 2, 2024, across major databases, identified 344 studies; 29 were analyzed in depth, focusing on: 1) rPPG signal extraction and heart rate estimation, where deep learning improved accuracy; 2) fatigue detection, showing benefits of multimodal data fusion; 3) mental state monitoring, with machine learning classifying cognitive load and distraction; and 4) emotional state monitoring and dataset development, indicating a trend toward holistic driver assessment. While deep learning has improved rPPG signal extraction, challenges remain in consistent physiological metric detection under dynamic conditions. There is also a lack of diverse population representation, especially female drivers, in datasets. The review underscores the potential of AI-enhanced camera systems to improve road safety, emphasizing the need for diverse, multimodal data integration for comprehensive monitoring.

INDEX TERMS Automotive safety, deep learning, driver monitoring, machine learning, physiological signals, rPPG, signal processing.

I. INTRODUCTION

A. BACKGROUND AND MOTIVATION

The importance of monitoring driver cognitive states, such as attention and alertness, has become increasingly critical in ensuring road safety. Traffic accidents often result from suboptimal cognitive functioning, including inattention and drowsiness, which have been identified as the main contributing factors to poor driving performance and high accident rates [1]. In fact, a significant percentage of road accidents can be attributed to the psychophysical condition of the driver, encompassing aspects such as sleepiness, sleep disorders,

and emotional states such as anger or excitement [2]. Advanced Driver Assistance Systems (ADAS) and other in-vehicle technologies seek to improve safety by assessing and augmenting risky driving behaviors through real-time cognitive state monitoring [1].

Recently, there has been increasing interest in low-cost, non-contact and pervasive methods for monitoring drivers. Specifically, video cameras and deep learning are used to monitor driver physiological signals and vital signs such as heart rate (HR), oxygen saturation (SpO₂), respiration rate (RR), Blood (BP) and temperature (TEMP). In turn, those signals can be used to analyze the driver physical, physiological and psychological state. This state-of-the-art intelligent driver monitoring solution have been driven by

The associate editor coordinating the review of this manuscript and approving it for publication was Yiming Tang¹.

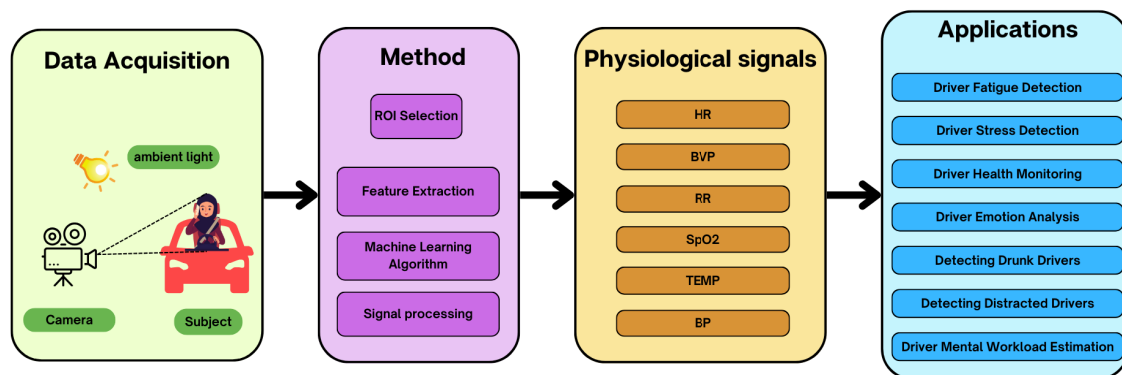


FIGURE 1. Block diagram of camera-based physiological measurement used in driver monitoring applications.

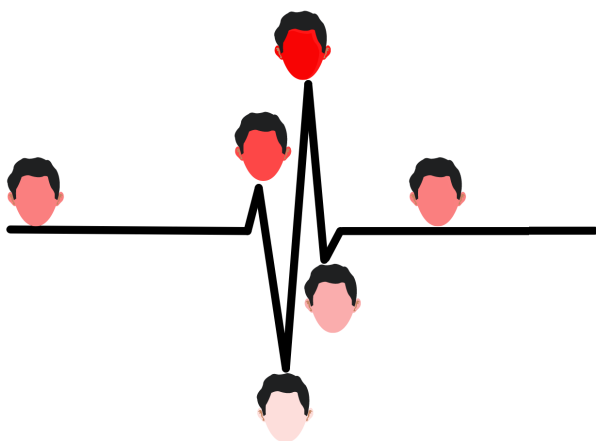


FIGURE 2. Diagram of rPPG Working Principle.

latest advancement in deep learning algorithms developed to extract photoplethysmography (PPG) signals remotely from video captured under ambient light, which is known as Remote photoplethysmography (rPPG)(see Fig. 1).

Figure 2 demonstrates the working principle of Remote photoplethysmography (rPPG). It operates by using video cameras to detect subtle variations in skin color caused by blood flow. As blood is pumped through the body, it causes slight changes in skin color with each heartbeat. These color changes are usually imperceptible to the human eye but can be captured by video cameras. The rPPG technology applies deep learning algorithms to analyze these minute color shifts, extracting vital physiological signals such as heart rate and oxygen saturation from the video data.

Several imaging devices can be employed in driver monitoring systems to capture these physiological signals. Each device offers unique advantages and challenges, depending on the specific monitoring requirements and environmental conditions. A summary of these devices, along with their principles, advantages, and disadvantages, is presented in Table 1.

The integration of rPPG into vehicle environments presents a unique and valuable opportunity. rPPG technology allows

the non-contact measurement of vital physiological signals, such as heart rate and oxygen saturation, using camera imaging technology. This method is particularly suited for the automotive setting because it is unobtrusive and does not require the driver to wear any special equipment, thus avoiding any additional distraction [3].

However, the need for a comprehensive review of unobtrusive driver monitoring through a camera-captured physiological metrics is underscored by the unique and challenging context in which these technologies are applied. The in-car environment introduces specific difficulties not present in more controlled settings like hospitals or homes and can be summarized as follows:

- **Variability in Illumination:** Lighting conditions inside a vehicle can vary dramatically due to changes in external light sources, affecting the visibility of physiological indicators.
- **Vehicle Motion:** The movement of the vehicle introduces complexity, disrupting the camera's focus or altering the frame of reference. The effect can be seen clearly in Fig. 3, where the same subject has a different rPPG signal depending on the driving location.
- **Context-Dependent Challenges:** Each driving scenario might present unique challenges, from the driver wearing sunglasses to changes in posture or facial expressions influenced by the driving context.

B. OBJECTIVE AND SCOPE

The dynamic environment inside a moving vehicle, characterized by varying lighting conditions, driver movement, and background noise, presents significant hurdles to accurately capturing and analyzing physiological signals. These factors can severely impact the reliability and accuracy of data derived from cameras, which are crucial for assessing driver well-being, attention, and fatigue levels in real-time. These challenges necessitate the development of unique machine learning algorithms and specialized devices designed to effectively handle the complexities of in-vehicle monitoring [2], [3].

TABLE 1. Summary of principle, advantages, and disadvantages of commonly used optical sensors for the unwired and unobtrusive physiological signal measurement.

Sensor Type	Underlying Mechanism	Benefits	Limitations
RGB imaging	Detect and analyze the body movement	Easy and burden-free measurement; cost-effective	Strongly affected by minor movements and lighting conditions
Thermal imaging	Measures emitted energy to assess temperature variations in the human body	Less affected by changes in lighting; operates effectively in complete darkness	Lacks geometric and texture details in captured images; relatively high hardware cost; impacted by ambient temperatures from surrounding surfaces
Hyperspectral imaging	Captures images across a wide range of continuous spectra from visible to infrared wavelengths	Offers high spatial and spectral resolution	Requires complex imaging equipment and extensive computation; slow data acquisition and analysis; limited time resolution; highly sensitive to motion artifacts
Depth imaging	Generates depth maps using time-of-flight or structured light technology	Provides precise depth information of the human body; functions well in low-light conditions	Easily obstructed by clothing or blankets; affected by sunlight; unsuitable for outdoor environments
Multimodal imaging	A combination of optical imaging sensors	Offers high sensitivity and resolution; provides more detailed and complementary biological data	Expensive with complex hardware; requires intricate data preprocessing algorithms

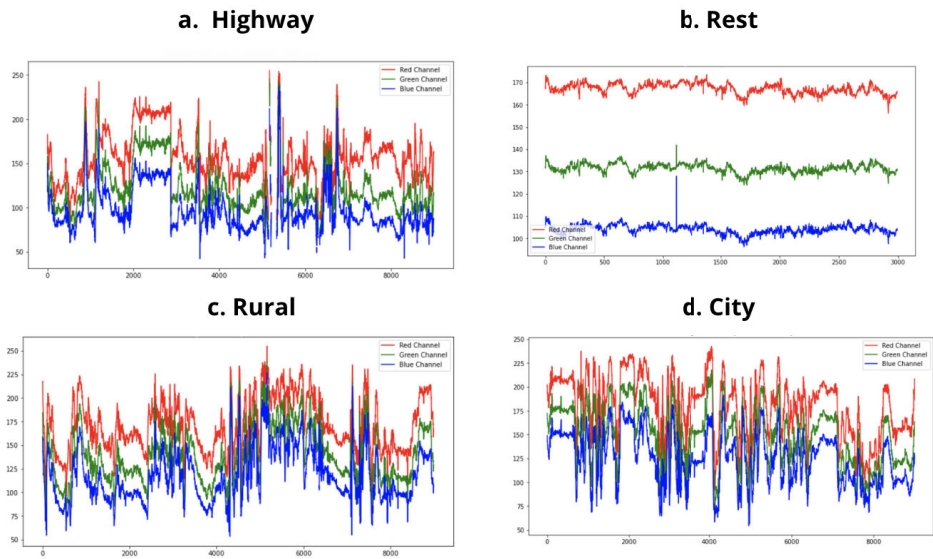


FIGURE 3. rPPG Signal for Subject 2209 in four different driving scenarios. The signals were acquired from In-Vehicle Health Monitoring Dataset [4]. (a) Highway (b) Rest (c) Rural (d) City.

Moreover, the need for such technology in cars is accentuated by the substantial time individuals spend in vehicles, often under stressful or fatiguing conditions [5]. With the increasing adoption of semi-autonomous and autonomous vehicles, the potential for cognitive disengagement from driving tasks grows, making effective monitoring even more crucial [5]. The implementation of rPPG technology can serve as a critical component in next-generation driver monitoring systems, providing continuous health monitoring without infringing on the comfort or privacy of the driver [3].

Studies have also highlighted that traditional fatigue monitoring methods are less effective under real-world driving conditions, often failing to capture the complex interplay of factors that contribute to driver fatigue. This underscores

the need for innovative approaches like rPPG that can offer more sensitive and accurate readings of physiological states [6], [7].

Given these challenges, this review is essential to:

- **Highlight Innovative Solutions:** Presenting how current research and technological advancements are overcoming these obstacles. This includes innovations in image processing algorithms, machine learning models capable of handling variability in data, and advancements in camera technology.
- **Identify Gaps in Research:** Pinpointing areas where further investigation is needed to develop more robust systems capable of accurate monitoring under a wide range of conditions.

- **Set a Foundation for Standardization:** Discussing methodologies and metrics for evaluating the performance of camera-based monitoring systems, paving the way for standardizing practices that ensure reliability and accuracy across different systems and vehicles.
- **Encourage Multidisciplinary Approaches:** Fostering collaborations across fields such as computer vision, automotive engineering, human factors, and artificial intelligence to create holistic solutions that address the multifaceted challenges of in-vehicle monitoring.
- **Real-Time Detection Imperatives:** Another critical aspect of implementing camera-based monitoring systems is the necessity for real-time detection and analysis. The ability to promptly recognize signs of fatigue, distraction, or health emergencies directly impacts the system's effectiveness in preventing accidents and enhancing driver safety. Developing algorithms and processing techniques that can operate efficiently in the vehicle's computing environment is crucial for achieving timely interventions.

Given these multifaceted challenges, this review aims to not only highlight innovative solutions and research gaps but also to encourage the development of robust, privacy-conscious, and real-time responsive monitoring systems. Such advancements are essential for the next generation of driver assistance technologies to truly enhance road safety and driver well-being.

C. STRUCTURE OF THE REVIEW

In the pursuit of enhancing driver safety and monitoring systems through technology, this systematic literature review seeks to explore the integration of machine learning (ML) with remote photoplethysmography (rPPG) for real-time driver monitoring. As we delve into this intersection of biometric sensing and artificial intelligence, our research questions are structured around three critical themes: 1) the development and performance of machine learning algorithms, 2) applications of rPPG in driver state monitoring, and 3) the identification of gaps in the existing research and directions for future technological advancements. By systematically analyzing how ML can optimize and enhance the functionality of rPPG systems in monitoring drivers, this review intends to contribute valuable insights into the evolution of safety mechanisms in the automotive industry.

1) Theme 1: Advancements and Performance of Machine Learning Algorithms in rPPG

- What machine learning algorithms have been developed for interpreting rPPG data for the purpose of driver monitoring, and how do they perform in terms of accuracy and reliability?
- Which machine learning models provide the best balance between performance and computational efficiency for use in real-time driver monitoring systems?
- How can machine learning algorithms be optimized for rPPG data collected in non-ideal

circumstances, such as varying lighting conditions and movement?

- What types of features are most effective when extracted from rPPG data for the purpose of machine learning-based driver monitoring?

2) Theme 2: Applications of rPPG in Driver State Monitoring

- How effective are machine learning-based rPPG systems in detecting driver fatigue and stress compared to systems using traditional physiological monitoring technologies?
- What advancements in machine learning could potentially enhance the predictive accuracy and real-time analysis capabilities of rPPG systems for driver monitoring?

3) Theme 3: Challenges and Future Directions

- What are the challenges and limitations of using ML techniques in rPPG systems for real-time driver monitoring under different driving conditions?
- What are the identified research opportunities and gaps?

II. RELATED WORK

Table 2 provides a comprehensive contrast between our systematic literature review on unobtrusive driver monitoring using physiological metrics and prior works. Our comparison spans several dimensions, including sensing modality, tracked parameters, machine learning techniques, and the intended purpose of each study.

Previous reviews, such as those by Lohani et al. [1], have delved into psychophysiological measures to assess drivers' cognitive states, with a focus on the significance and scope of various indices in real-world driving scenarios. Our review, however, extends beyond these standard physiological metrics by integrating both deep learning (DL) and shallow learning (SL) methodologies. This allows for a more nuanced understanding of drivers' states, enhancing both predictive and real-time analysis capabilities.

Zhang et al. [7] explored the progression from data acquisition to fatigue assessment. Our work builds upon this by integrating recent advancements in deep learning, thereby improving both the accuracy and responsiveness of fatigue monitoring systems.

Razak et al. [2] and Sikander and Anwar [6] discussed physiological-based monitoring and fatigue detection technologies, emphasizing the current technological landscape. In contrast, our review updates these discussions to reflect the latest innovations up to 2024 and critically examines their integration within systems aimed at enhancing driver safety and health.

Wang et al. [5] presented a comprehensive view on in-vehicle health monitoring. Our review distinguishes itself by not only discussing the sensor technology but also exploring how these technologies are integrated into existing systems

to optimize both driver performance and health outcomes in real-time scenarios.

The foundational work of Leonhardt et al. [3] and Liu et al. [8] on unobtrusive monitoring and fatigue detection technologies laid the groundwork for this field. Our review advances this discussion by intertwining these technologies with current machine learning algorithms, providing insights into the future trajectory of unobtrusive driver monitoring systems that leverage both physiological and behavioral data for enhanced predictive accuracy.

Freitas et al. [9] conducted a systematic review of electrocardiogram (ECG) and heart rate monitoring techniques for detecting fatigue and drowsiness in drivers. While their work highlights the potential of physiological metrics, it primarily focuses on contact-based sensing modalities. In contrast, our review uniquely concentrates on contactless sensing technologies, offering a more inclusive evaluation of modern unobtrusive monitoring methods that integrate both physiological and behavioral data.

Al-Quraishi et al. [10] explored wearable and non-wearable sensor technologies for driver fatigue detection, emphasizing the advancements in sensor implementations and detection algorithms. Their review provides a broad overview of sensor technology but does not delve into the exclusive use of contactless modalities or the integration of behavioral monitoring. Our work bridges this gap by addressing both physiological and behavioral monitoring in contactless setups, offering insights into how these approaches can be optimized for real-time safety and health applications.

By focusing on contactless sensing methods, our review diverges from prior works that predominantly rely on contact-based systems or general sensor technologies. We also highlight both deep learning and shallow learning methodologies in advancing unobtrusive monitoring systems. This emphasis on contactless sensing and the combined use of physiological and behavioral data positions our review as a significant contribution to the evolving field of driver monitoring.

The majority of prior reviews explore both contact and contactless sensing methods to monitor drivers, offering a comprehensive approach to data collection [1], [2], [3], [5], [6], [7]. Regarding the utilization of machine learning, studies such as those by Zhang et al. [7], Razak et al. [2], Liu et al. [8], and Sikander and Anwar [6] have incorporated both deep learning (DL) and shallow learning (SL) techniques in their driver monitoring systems. In contrast, some reviews like those by Wang et al. [5] and Leonhardt et al. [3] did not discuss the specific machine learning approaches employed in the studies they reviewed. In terms of tracking both physiological and behavioral parameters, the studies by Lohani et al. [1], Zhang et al. [7], Wang et al. [5], and Sikander and Anwar [6] provide a multi-faceted perspective by examining both data types, ensuring a thorough understanding of driver states.

Compared to these studies, our review exclusively focuses on contactless sensing methods, which represent a significant advancement in the field by reducing intrusiveness and enhancing driver comfort. Furthermore, our work highlights the utilization of both deep and shallow learning techniques, as well as the examination of both physiological and behavioral parameters. This dual approach positions our study as a more holistic, recent, and advanced contribution to the literature on unobtrusive driver monitoring systems.

III. METHOD

This systematic review adheres to the updated guidelines provided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement [11].

A. ELIGIBILITY CRITERIA

In our systematic review, studies were selected based on stringent eligibility criteria adhering to the PICO framework. Table 3 outlines our eligibility criteria in a structured format, aligned with the well-recognized PICO model, which includes Population, Intervention, Comparator, Outcome, and an additional component for Study Characteristics.

B. DATA SOURCE AND SEARCH STRATEGY

A thorough and systematic search was conducted across four electronic databases to gather relevant studies for this review. Specifically, the search was performed in Scopus, IEEE Xplore, ScienceDirect, and Web of Science to maximize the coverage of the literature in the fields of physiological metric estimation via facial video analysis in automotive settings. The search strings were tailored to each database's specific syntax and capabilities. The number of documents retrieved from each database, as listed in Table 4, reflects the scope and yield of the search strategy.

C. DATA COLLECTION, STUDY SCREENING, AND SELECTION

Data retrieved from the four databases were downloaded directly in the structured RIS file format. The records obtained, which include titles, abstracts, authors, journals, funding information, and more, were imported into Covidence, a software designed to manage and streamline systematic reviews. Covidence facilitated the removal of duplicates, and the initial screening processes, such as title and abstract review, followed by full-text screening.

Subsequent to this, researcher Soha G. Ahmed (SGA) meticulously performed the screening of titles, abstracts, and full texts. To ensure the review process was comprehensive and to address any doubts or ambiguities that emerged during the screening, a senior reviewer, Nazar Zaki (NZ), provided his expertise. This collaborative method strengthened the screening process, substantially reducing the likelihood of oversight and reinforcing the selection phase's stringency.

TABLE 2. Content covered by recent reviews on unobtrusive driver monitoring through physiological metrics.

Ref	Sensing		Tracked Parameters ¹			ML ²		Purpose		Time
	Contact	Contactless	Env.	Physio.	Behav.	DL	SL	Safety	Health	
[1]	✓	✓		✓	✓			✓		before 2019
[7]	✓	✓		✓	✓	✓	✓	✓		before 2022
[2]	✓	✓		✓		✓	✓	✓		2018 - 2022
[5]	✓	✓	✓	✓	✓			✓	✓	2009–2019
[3]	✓	✓		✓					✓	before 2018
[8]		✓		✓		✓		✓		before 2022
[6]	✓	✓		✓	✓	✓	✓	✓		before 2018
[9]	✓	✓		✓		✓	✓	✓		2018–2023
[10]	✓	✓		✓	✓	✓		✓		2014–2024
Ours		✓		✓	✓	✓	✓	✓	✓	2014–2024

¹ Tracked Parameters: Env. = Environmental, Physio.=Physiological, Behav.= Behavioral.

² ML:Machine learning; DL=Deep Learning, SL=Shallow Learning.

TABLE 3. Eligibility criteria for the selection of studies in the systematic review following PICO framework.

Criteria	Inclusion	Exclusion
Population	Studies with populations within automotive environments (real cars, simulators, lab simulations).	Studies focusing on non-automotive environments.
Intervention/ Exposure	Studies applying data mining, knowledge discovery, deep learning, neural networks, or machine learning for facial video-based physiological metrics estimation, and assessing additional parameters like eye blink and facial expression.	Studies not concerning facial video-based physiological metrics estimation. Studies that do not use data mining for physiological metrics estimation.
Comparator/ Context	Prestigious international conference papers (CVPR, ICCV, ECCV). Studies using reference devices or standard methods for comparison of physiological metrics.	Studies lacking a clear benchmark or comparison for the interventions/exposures. General conference papers and book chapters, except for specified conferences.
Outcome	Studies that measure the effectiveness, accuracy, or other relevant outcomes of the specified data analytical techniques in automotive settings.	Studies not measuring outcomes related to these specific data analytical techniques or not related to automotive settings.
Study Characteristics	Peer-reviewed journal papers published within the last ten years. Articles published in English.	Conference papers and book chapters (general exclusion, with exceptions noted above). Patents and non-peer-reviewed literature. Studies outside the specified date range. Articles published in languages other than English.

D. QUALITY ASSESSMENT & DATA SYNTHESIS

Quality assessment in our review is pivotal, given the application of remote photoplethysmography (rPPG) technology in dynamic automotive environments. We assess the methodological integrity and consistency of the included studies, focusing on the accuracy and reliability of RGB camera-captured physiological metrics under varied automotive conditions. Criteria for assessment include the robustness of camera technology against environmental variables such as lighting changes, driver movement, and external vibrations, as well as the sophistication of algorithms used to interpret physiological data.

Data synthesis in this review involves combining results from various studies to address the efficacy and challenges of implementing rPPG systems in vehicles. We adopt a narrative synthesis approach, considering the heterogeneity of the studies in terms of experimental setups, camera

technology, and machine learning algorithms. This synthesis will highlight common findings related to the accuracy and reliability of physiological measurements obtained in-vehicle, discussing how different setups impact data quality. We also explore advancements in machine learning models that enhance data interpretation, aiming to draw conclusions about the current state of the art and areas requiring further research.

This detailed synthesis not only aids in understanding the current technological landscape but also in identifying gaps in technology and methodology that could be addressed by future research, thus paving the way for more sophisticated and unobtrusive driver monitoring systems.

IV. RESULTS

The results section is systematically structured to facilitate a comprehensive understanding of the research findings. It is divided into three distinct subsections, each tailored to address different facets of the study.

The initial subsection provides a synthesized overview of the study selection process and the screening outcomes, setting the stage for a detailed analysis of the gathered data.

Following this, we delve into the evolution and performance of machine learning algorithms within the automotive context. It contrasts the efficiency of these contemporary techniques with traditional monitoring systems and assesses their integration with benchmark datasets specifically curated for rPPG signal extraction in vehicles.

Finally, the focus shifts to the practical implications of rPPG technology, particularly its application in cars for monitoring driver states, such as fatigue and emotional well-being. This part will underscore the real-world applications and benefits of rPPG in enhancing driver safety and wellbeing.

A. SCREENING RESULTS AND STUDIES OVERVIEW

1) DESCRIPTIVE STATISTICS OF INCLUDED STUDIES

Out of 344 articles retrieved from the search strategy presented in Table 4 using multiple databases, 29 articles were included, followed by data collection and analysis.

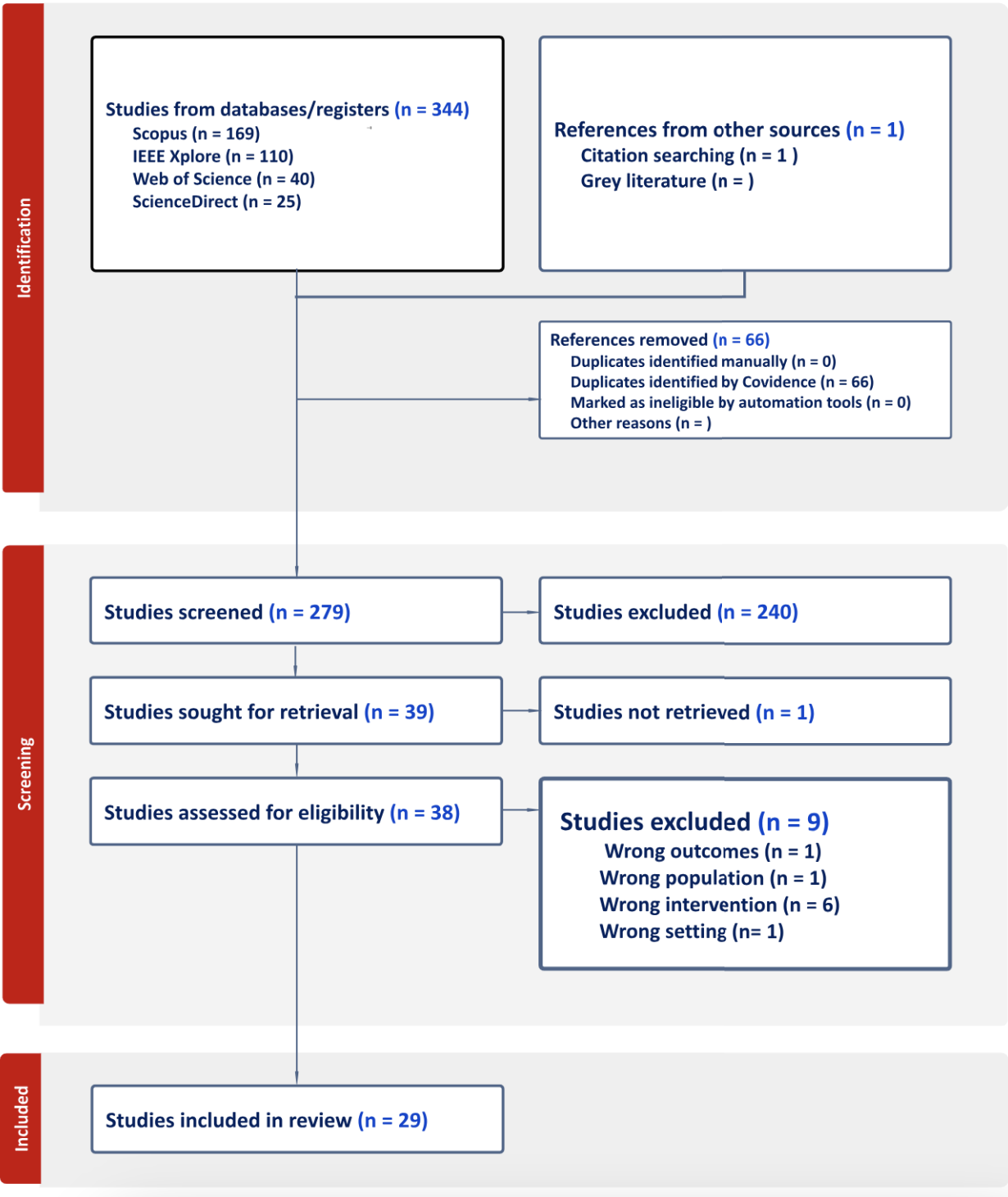


FIGURE 4. PRISMA Flow Diagram for this systematic review.

Fig. 4 depicts the PRISMA flow diagram illustrating the article’s screening process.

2) CHARACTERISTICS OF SELECTED STUDIES

Table 5 below summarizes the studies included in this review. Each study is categorized by its main purpose,

the dimension of focus—whether mental, physical, physiological, or emotional-, the year of publication (Fig. 5b) and the journal. Fig. 5a shows that the majority of the included studies focus on the physiological aspect of driver monitoring - such as rPPG signal extraction,-with more than 34% of the included studies. Fig 5c illustrates the number of

TABLE 4. Boolean search strings and the number of documents retrieved from each database.

Database	Boolean Search Strings	No. of doc
Scopus	TITLE-ABS-KEY (('data mining' OR 'knowledge discovery' OR 'deep learning' OR 'neural networks' OR 'machine learning') AND ('RGB cameras' OR 'color camera' OR 'video-based monitoring' OR 'non-contact' OR 'contactless' OR 'imaging' OR 'camera' OR 'video') AND ('physiological metrics' OR 'biometrics' OR 'vital signs' OR 'heart rate' OR 'remote PPG' OR 'iPPG' OR 'remote photoplethysmography' OR 'PPGi' OR 'respiration' OR 'breathing' OR 'heart rate variability' OR 'blood oxygen saturation' OR 'blood pressure') AND ('vehicle' OR 'car' OR 'driving' OR 'driver' OR 'driver monitoring system' OR 'driver state detection' OR 'in-vehicle monitoring')) AND PUBYEAR > 2013 AND PUBYEAR < 2025	169
IEEE Xplore	((('data mining' OR 'knowledge discovery' OR 'deep learning' OR 'neural networks' OR 'machine learning') AND ('RGB cameras' OR 'color camera' OR 'video-based monitoring' OR 'non-contact' OR 'contactless' OR 'imaging' OR 'camera' OR 'video') AND ('physiological metrics' OR 'biometrics' OR 'vital signs' OR 'heart rate' OR 'remote PPG' OR 'iPPG' OR 'remote photoplethysmography' OR 'PPGi' OR 'respiration' OR 'breathing' OR 'heart rate variability' OR 'blood oxygen saturation' OR 'blood pressure') AND ('vehicle' OR 'car' OR 'driving' OR 'driver' OR 'driver monitoring system' OR 'driver state detection' OR 'in-vehicle monitoring') AND (Year:"2014" OR Year:"2015" OR Year:"2016" OR Year:"2017" OR Year:"2018" OR Year:"2019" OR Year:"2020" OR Year:"2021" OR Year:"2022" OR Year:"2023" OR Year:"2024"))	110
Science Direct	('deep learning' OR 'machine learning') AND ('RGB cameras' OR 'video-based monitoring') AND ('vital signs' OR 'heart rate variability') AND ('vehicle' OR 'driving')	25
Web of Science	TS=((('data mining' OR 'knowledge discovery' OR 'deep learning' OR 'neural networks' OR 'machine learning') AND TS=('RGB cameras' OR 'color camera' OR 'video-based monitoring' OR 'non-contact' OR 'contactless' OR 'imaging' OR 'camera' OR 'video') AND TS=('physiological metrics' OR 'biometrics' OR 'vital signs' OR 'heart rate' OR 'remote PPG' OR 'iPPG' OR 'remote photoplethysmography' OR 'PPGi' OR 'respiration' OR 'breathing' OR 'heart rate variability' OR 'blood oxygen saturation' OR 'blood pressure') AND TS=('vehicle' OR 'car' OR 'driving' OR 'driver' OR 'driver monitoring system' OR 'driver state detection' OR 'in-vehicle monitoring')) AND PY=(2014-2024)	40

publications within each identified journal. 'Sensors' leads with the highest count, followed by 'IEEE Access' indicating a prominent source of research in this domain. The journals listed further highlight the interdisciplinary nature of the research, spanning fields from biomedical engineering to intelligent transportation systems. This detailed aggregation of studies serves as the foundation for our systematic analysis and discussion on the efficacy and challenges of integrating ML with rPPG technologies in driver monitoring systems. The specific purposes of the studies, which reflect various aspects of driver state monitoring, are summarized as follows:

- **Driver Fatigue:** This refers to a decline in mental and/or physical performance due to prolonged driving, lack of sleep, or extended periods of mental or physical activity. Fatigue leads to slower reaction times, decreased vigilance, and impaired decision-making [27].
- **Driver Drowsiness:** Often a subset or a symptom of fatigue, drowsiness specifically pertains to the feeling of being sleepy while driving. This state can result from inadequate rest but is characterized more by yawning, heavy eyelids, and drifting thoughts [23].
- **Drunk Driving:** This involves operating a vehicle under the influence of alcohol. Alcohol impairs cognitive abilities and reaction times, increasing the risk of accidents significantly. It's legally regulated in most places [22].
- **Driver Stress:** This is an emotional or mental strain experienced while driving, often triggered by traffic conditions, personal or work pressures, or aggressive driving behaviors. Stress can affect concentration and increase irritability, impacting driving performance [16].
- **Driver Distractions:** These are factors that divert a driver's attention from the road. Distractions can be external (e.g., billboards) or internal (e.g., mobile phone use, passengers), leading to reduced awareness of the driving environment and higher accident risk [17].
- **Driver's Cognitive Load:** This term specifically refers to the amount of mental processing power required to perform driving-related tasks. Cognitive load can vary

based on the complexity of the driving environment and the tasks the driver is engaged in, such as navigating, responding to traffic signals, or making quick decisions. Cognitive load is typically discussed in the context of how information is processed and managed by the brain. Higher cognitive loads can lead to errors or slower reaction times because the brain's capacity to process information is finite [19].

- **Drivers' Mental Workload Levels:** This term is broader and includes not only the cognitive aspects but also the emotional and psychological stress that can affect a driver. Mental workload encompasses the overall effort and resources used by a driver to meet the demands of driving. It includes managing distractions, fatigue, and stress, in addition to the cognitive processing of the driving task itself. Measuring mental workload often involves assessing how much capacity the driver has left to respond to unexpected events or additional tasks [18].

Fig. 6 illustrates the distribution of the 29 studies included in the review, categorized by the specific driver monitoring characteristics they address, such as heart rate monitoring, cognitive load classification, and driver emotion recognition.

B. ADVANCEMENTS AND PERFORMANCE OF MACHINE LEARNING ALGORITHMS IN rPPG

Machine learning algorithms have been integral to the development and enhancement of rPPG systems, with each algorithm contributing unique approaches to the extraction and analysis of physiological signals. The included studies in the review reveal a diverse array of methodologies that have been applied to driver monitoring. Figure 7 portrays the machine learning algorithms landscape within the included studies. It delineates how these algorithms are categorized and provides insights into the breadth and depth of the research undertaken.

Figure 8 quantifies the presence of different machine learning techniques across the reviewed literature. It underscores the traction deep learning has gained in recent times while also acknowledging the continuing relevance of traditional

TABLE 5. Overview of Included Studies on the review, detailing each study's purpose, focus dimension, publication year, and journal.

Ref.	Purpose	Dimension	Year	Journal
[12]	Dataset		2022	Data MDPI
[13]	Dataset		2023	Sensors
[14]	Driver Emotion Recognition	Emotional	2021	IEEE Transactions on Intelligent Transportation Systems
[15]	Driver Emotion Recognition	Emotional	2021	Biomedical Signal Processing and Control
[16]	Driver Stress	Emotional	2024	Traffic Injury Prevention
[17]	Driver Distractions	Mental	2020	IEEE Access
[18]	Driver Mental Workload Levels	Mental	2022	Sensors
[19]	Driver's Cognitive Load Classification	Mental	2021	Sensors
[20]	Driver's Cognitive Load Classification	Mental	2020	Biomedical Signal Processing and Control
[21]	Takeover Performance Prediction	Mental	2023	IEEE Communications Magazine
[22]	Drink Driving	Mental, Physical	2022	IEEE/ACM Transactions on Networking
[23]	Driver Drowsiness	Mental, Physical	2019	Accident Analysis and Prevention
[24]	Driver Drowsiness	Mental, Physical	2024	Lecture Notes of the ICST
[25]	Fatigue Detection	Physical	2024	Sensors
[26]	Fatigue Detection	Physical	2021	IEEE Transactions on Intelligent Transportation Systems
[27]	Fatigue Detection	Physical	2021	IEEE Access
[28]	Fatigue Detection	Physical	2024	Lecture Notes in Computer Science
[29]	Fatigue Detection	Physical	2022	IEEE Transactions on Intelligent Transportation Systems
[30]	Fatigue Detection	Physical	2020	IEEE Access
[31]	Heart Rate Monitoring	Physiological	2019	IEEE Access
[4]	Heart Rate Monitoring	Physiological	2023	Scientific Reports
[32]	Heart Rate Monitoring	Physiological	2024	Sensors
[33]	Heart Rate Variability Estimation	Physiological	2022	IEEE Internet of Things Journal
[34]	Respiratory Rate Estimation	Physiological	2023	Lecture Notes in Networks and Systems
[35]	Respiratory Rate Estimation	Physiological	2023	Scientific Reports
[36]	rPPG	Physiological	2023	IEEE Transactions on Instrumentation and Measurement
[37]	rPPG	Physiological	2022	IEEE Transactions on Instrumentation and Measurement
[38]	rPPG	Physiological	2022	IEEE Transactions on Instrumentation and Measurement
[39]	rPPG	Physiological	2021	Intelligent Automation and Soft Computing

machine learning techniques. It also illuminates the tendency to utilize deep learning for rPPG signal extraction and traditional machine learning for driver state monitoring within the reviewed studies.

The systematic review reveals a clear trend in the utilization of machine learning within the realm of rPPG systems for driver monitoring. Deep learning algorithms have been predominantly utilized for developing new methods and enhancing the accuracy of rPPG signal extraction. In contrast, traditional machine learning algorithms have been applied more rigorously to interpret these signals, particularly for tasks centered around monitoring specific driver states such as fatigue and emotion. This subsection aims to dissect and elucidate the distinct roles these two branches of machine learning occupy, shedding light on their respective applications as supported by the literature.

An exploration into the evolution of these algorithms and their applications in rPPG technologies is presented, charting their progress and summarizing their deployment within the studies reviewed. The performance of these machine learning approaches is critically evaluated, assessing their capabilities and effectiveness in overcoming the intrinsic challenges of rPPG signal extraction and analysis within driver monitoring contexts.

Further scrutiny will be applied to the efficacy of deep learning algorithms in their pivotal role of extracting high-fidelity rPPG signals. Concurrently, we will explore the capacity of traditional machine learning algorithms to utilize these signals effectively, meeting the diverse objectives of driver state monitoring. This section aims to highlight the synergy between these methodologies, positing that their combined use could significantly enhance the reliability and practicality of driver monitoring systems.

The discussions that follow will delve into the strengths and limitations of these techniques, reflecting on their individual contributions to the field. Additionally, the review will suggest prospective avenues for future research, potentially catalyzing further advancements and innovations in the technology of non-contact physiological monitoring for drivers.

1) EVOLUTION AND PERFORMANCE OF MACHINE LEARNING ALGORITHMS FOR rPPG

Table 6 compares the different experimental setups across the studies that collected their own dataset or used public datasets to develop a rPPG Signal extraction algorithm or improve the performance of existing algorithms. Table 7 presents a comparison between those studies in terms of the machine learning algorithm used, the Region of Interest (ROI), features extracted, and evaluation metrics used.

In a study by [34], the results showed that the real-time heart rate detection system had good results but was affected by motion. Another study by [38] demonstrated that deep learning yielded better results than traditional approaches, with an error compensation framework reducing the MAE to below 10 BPM, showing significant improvements over state-of-the-art methods, especially in challenging cases. The research by [37] revealed that fusion models had improved performance compared to previous methods, with a significant improvement in RMSE when using two or more transformers, and the framework demonstrated stable HR estimation performance. In the study by [32], the models achieved better results compared to previously published works on the LGI-PPGI and V4V datasets. In another study by [31], the results indicated that the performance of the

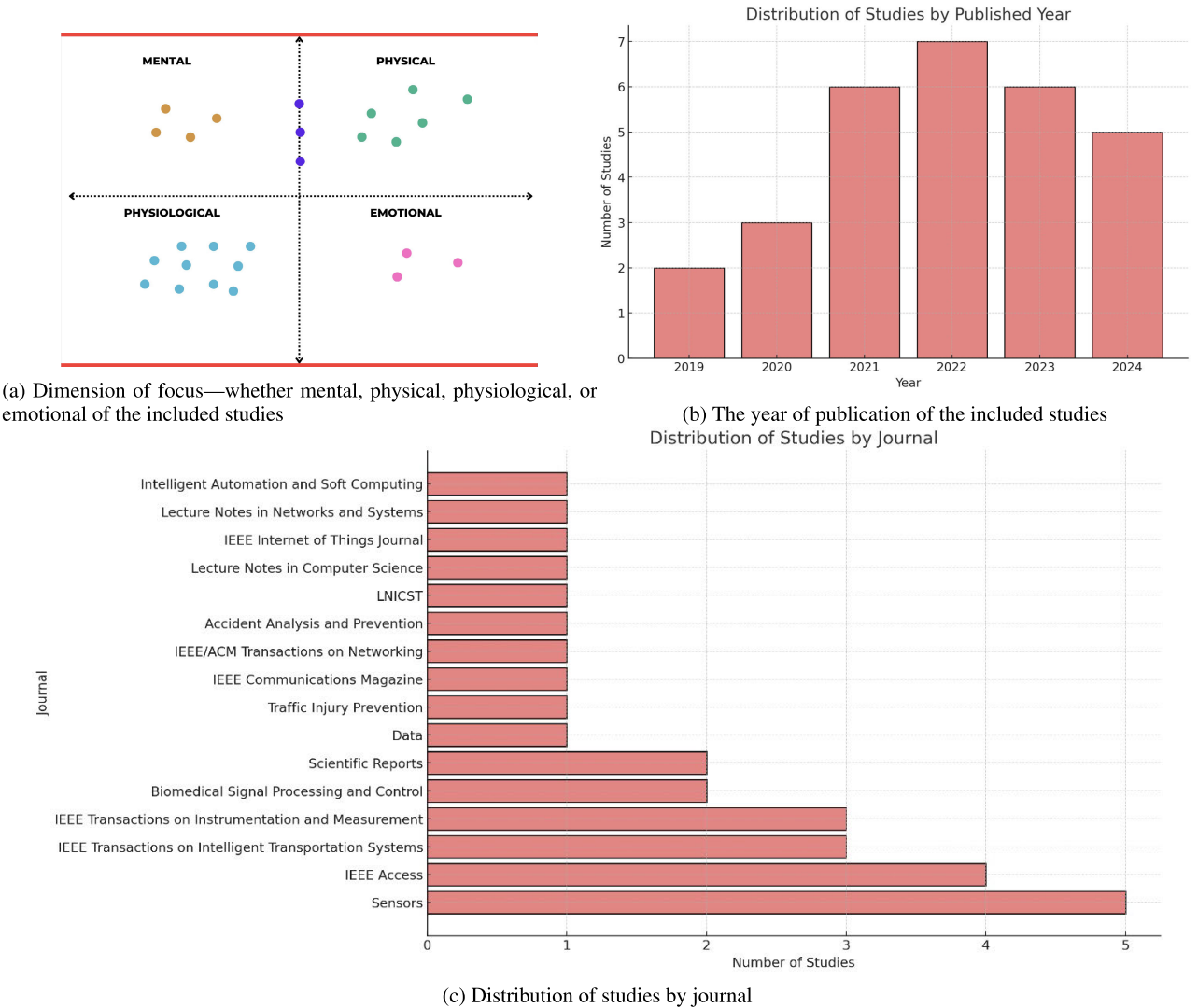


FIGURE 5. Characteristics of the included studies.

proposed method was better in overcast and rainy weather compared to sunny weather. The research by [39] showed that the proposed algorithm was able to accurately extract physiological parameter information and facial expressions from real-time video sequences, tested on five volunteers in a road experiment, with a good convergence speed and short training time, and validated by combining facial expression recognition with physiological parameter monitoring, accurately evaluating the driver’s physical state. In the study by [35], a performance difference for respiratory rate detection between different driving scenarios was found, with a difference of 0 to 3 breaths per minute. The research by [4] showed that the hybrid signal fusion of three signals outperformed previous results for heartbeat detection under real driving conditions, but iPPG’s performance was worse than ECG and PPG. Finally, in the study by [33], the proposed approach did not generalize well to the leave-one-subject-out setting due to a limited sample size of only nine drivers, and a larger-scale field study with more subjects is expected to

improve generalizability. The dataset comprised 686 hours of video recordings paired with HRV measurements, collected while participants drove over relatively long distances.

2) BENCHMARK DATASETS AND VALIDATION METHODS

The veracity and generalizability of machine learning models for rPPG are contingent upon the quality of the datasets used for training and validation. This subsection expounds on the benchmark datasets that have been publicly released, delineating their features, subject demographics, capture environments, and the standards used for validation. Such datasets are pivotal for developing robust models capable of performing under the multifarious conditions encountered in real-world driving scenarios. Table 8 encapsulates a summary of these critical datasets.

The Multimodal Dataset for Various Forms of Distracted Driving (2017) [40] includes comprehensive data captured in a driving simulator under different distraction conditions. It encompasses variables such as speed,

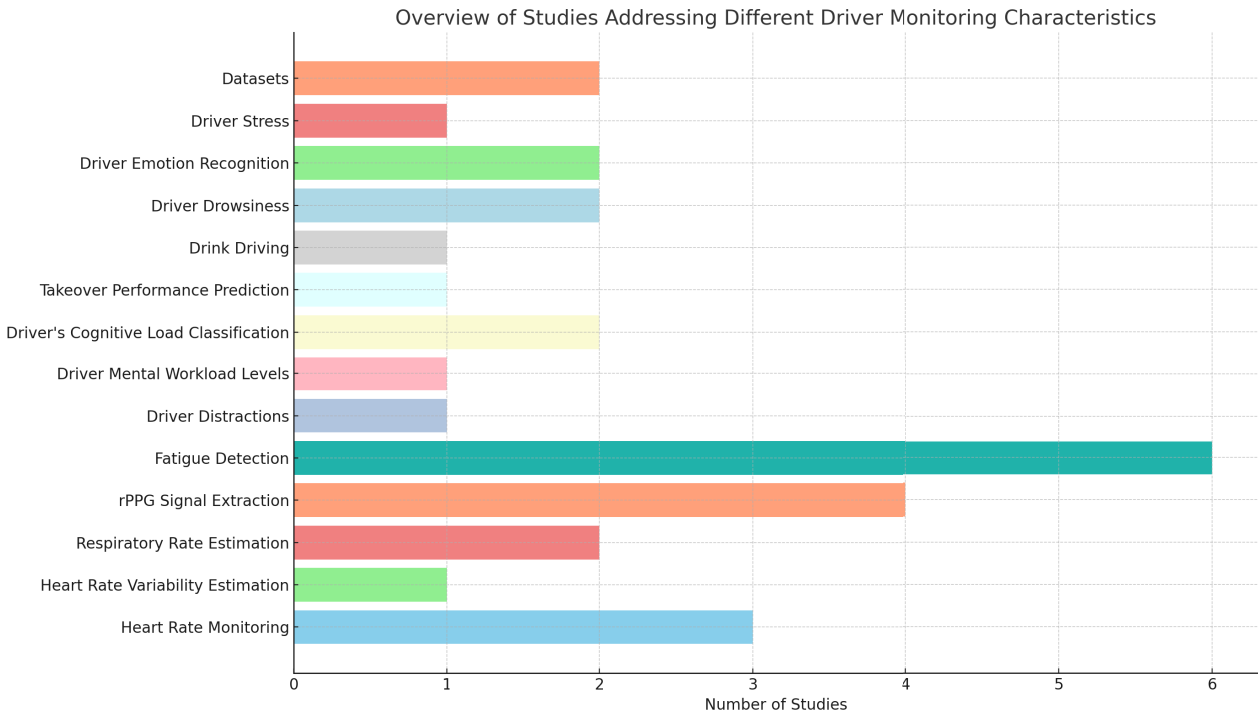


FIGURE 6. Overview of studies addressing different driver monitoring characteristics.

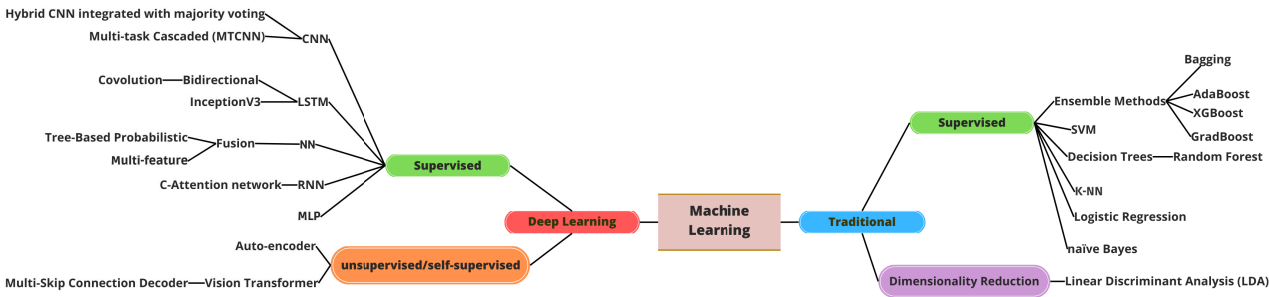


FIGURE 7. A systematic review-derived mind map categorizing machine learning algorithms reported in the literature, detailing the various approaches discussed in the included papers. The map organizes algorithms into categories of deep learning, including supervised, unsupervised, and self-supervised learning, and contrasts them with traditional machine learning methods.

acceleration, brake force, steering input, lane position, perinasal electrodermal activity (EDA), palm EDA, heart rate, and breathing rate, facial expressions, and eye tracking data. This dataset features 68 subjects (35 males and 33 females), with data recorded using a FireWire CCD monochrome zoom camera at 1024×768 pixels and 15 frames per second (FPS) and a Thermal camera Tau 640 at 640×512 pixels and 7.5 FPS. Physiological signals were measured using Shimmer3 GSR sensors for EDA and Zephyr BioHarness 3.0 for heart rate and breathing rate, all collected in a controlled lab environment using a driving simulator.

- **Focus:** This dataset captures driver behavior in a simulated environment under various distraction conditions. It includes comprehensive multimodal data such as physiological signals, driving performance metrics, and visual data from cameras.

- **Pros:** Ideal for studies investigating the impact of different distractions on driving behavior, analyzing physiological responses, and developing models for detecting distracted driving based on multiple input streams. It has a relatively balanced sample.
- **Cons:** The dataset was collected exclusively in a controlled lab environment using a simulator, which may not fully replicate the sense of danger or real-world driving conditions experienced during actual distractions.

The MERL-rice NIR Pulse (Driving) (2020) [41] dataset is notable for being the first publicly available video dataset with ground truth pulse signals captured during driving. It includes driving with different motions and various filters for ambient light. The dataset consists of 18 subjects (16 males and 2 females) across 185 videos. Data was recorded using an RGB camera at 640×640 pixels

Ordered Frequency of Machine Learning Techniques in rPPG Systems for Monitoring Drivers (Grouped by Category)

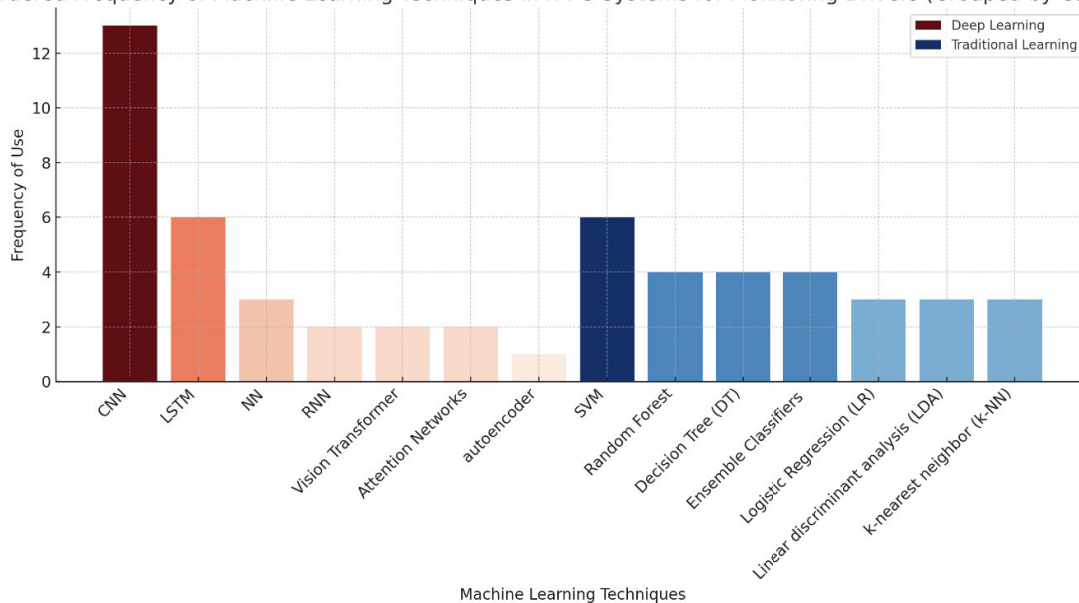


FIGURE 8. The ordered frequency of machine learning techniques in rPPG systems for monitoring drivers, categorized by their application in the research papers. Deep learning techniques are shown to dominate rPPG signal extraction, whereas traditional learning techniques are more often employed to analyze these signals for practical monitoring tasks such as fatigue and emotion detection.

and 30 FPS and an NIR camera at 640×640 pixels and 60 FPS. The ground truth PPG waveform was obtained using a CMS50D+ finger pulse oximeter recorded at 60 FPS, with data collected in a real-world driving environment.

- **Focus:** This dataset is unique in providing NIR and RGB video data along with ground truth pulse signals during real-world driving, focusing on capturing the physiological effects of driving under different lighting and motion conditions.
- **Pros:** Particularly useful for developing and testing algorithms for remote pulse detection in dynamic lighting environments typical of driving scenarios, enhancing models that require robustness against such variations.
- **Cons:** The sample is unbalanced, with a significantly low representation of female participants. Additionally, the use of a finger pulse oximeter as the ground truth device may not be ideal for real-world in-car scenarios, as it could introduce discomfort or impracticality during extended driving sessions.

The DriverMVT (2022) [12] dataset is designed for real-time monitoring of drivers, with videos collected in diverse driving scenarios. It includes metadata such as geographic coordinates, speed, and light conditions, captured using USB and smartphone cameras. This dataset includes 9 subjects (7 males and 2 females), with data recorded at 30 FPS using USB cameras and at 1080×1920 pixels and 60 FPS using smartphones. Heart rate data was measured using a Xiaomi Mi Band 3, with data collected in real-world driving conditions.

- **Focus:** Focuses on real-time monitoring of drivers using video data collected in real-world conditions,

accompanied by metadata like speed and geographic coordinates.

- **Pros:** This dataset supports the development of real-time driver monitoring systems that can adapt to various driving conditions and can be integrated with navigation and vehicle control systems for enhanced safety.
- **Cons:** The Xiaomi Mi Band 3 used for heart rate measurement is a consumer-grade wearable device, which may lack the precision required for rigorous research applications, especially in dynamic environments. Also, the sample has limited female representation.

The In-Vehicle Health Monitoring Dataset (2023) [4] features ECG, PPG, and RGB video for continuous health monitoring in various driving scenarios. It also incorporates face recognition for real-time face landmarks detection and medical analysis. This dataset includes 19 subjects (13 males and 6 females), with data recorded using an RGB camera at 10 FPS. Reference ECG, PPG, and steering wheel ECG signals were sampled at 500 Hz, collected in real-world driving scenarios in a Volkswagen Tiguan.

- **Focus:** Combines ECG, PPG, and video data for comprehensive health monitoring within a vehicle, emphasizing continuous data collection across different driving scenarios.
- **Pros:** Suitable for developing systems that integrate health monitoring with driving performance, potentially leading to applications that can alert drivers based on health data or adjust vehicle operation in response to detected health issues.
- **Cons:** The dataset has a limited sample size with an unbalanced gender representation, which may affect its

TABLE 6. Comparison of experimental setup across studies in rPPG signal extraction.

Ref	Subjects	Dataset	Camera	Ground truth	Environment	Conditions tested
[34]	not specified	authors' own dataset	monocular cameras(a smartphone camera) and a smartwatch to collect data	a smartwatch to collect data	real world environment, in the car, using a smartphone camera and a smartwatch to collect data	lighting conditions, driver movement, car movement
[38]		PURE dataset	Logitech C920 frame rate of 30 FPS and HD Pro camera	CMS50E for PPG	The study was conducted in both the real world and in simulated environments The training dataset was recorded in a sunny passenger environment and the validation dataset was recorded in a rainy passenger environment Additionally, two self-constructed datasets were recorded for compact car drivers and passengers with driving disturbance However, the PURE dataset was recorded in a lab setting	lighting (sunny and rainy), movement (head turns, car shaking, and light changes), and noise (with significant motion)
[37]	107 subjects	VIPL-HR and MR-NIRP-C ARDATASET	Logitech C310 web-camera, the front camera of HUAWEI P9 smartphone, and RealSense F200 camera .25 fps to 30 fps	CMS 50D finger pulse oximeter	The study was conducted in the real world, specifically in a dark low-light environment using a driving dataset	head movements and illumination variations
[36]	18 participants	MERL-rice NIR pulse Car and authors' own datasets:the Drivers and Driver-Highway datasets, and the Rainy and Sunny Passengers datasets	Logitech C920 with a frame rate of 30 FPS and the NIR camera with a frame rate of 60 FPS	The CMS50D+ finger pulse oximeter is used as the gold standard for physiological signals monitoring	The study was conducted in a real-world environment, specifically in a car while driving on a specific path in the city	insufficient light sources on rainy days at night, head movements, varying illumination and background, and night and rainy night scenes
[32]	179 (diverse racial groups ,age(18 and 66)	V4V, LGL-PPGI, and the author's own dataset	smartphone cameras, Logitech HD C270 webcams(frame rates : 60 FPS and 25 FPS for the smartphone and webcam cameras)	blood pressure monitor, oximeter, and electrocardiogram	The study was conducted in a real-world environment in a moving vehicle	different illumination and movement conditions, specifically in a moving vehicle
[31]	14 (11M, 3F),20-25years	HCI driver dataset	cameras built on TAIWAN ITS-2, bitmap format with VGA (640x480) resolution, 8-bit color depth, 30 fps during the day At night, the camera is switched to a de-IR-filter web camera with a reduced frame rate of 15 fps due to limitations of the camera	The Scosche Rhythm+ armband heart rate monitor	real-world, car(Mitsubishi SA VRIN)	rainy nights in the city, changing light at sunset, rainy and overcast days, and sudden luminance variations from driving through shade
[39]	5 volunteers		cameras: AONI A30 and a low-cost webcam(frame rate is 30 FPS, they are both USB cameras)	ECG (Electrocardiogram)	real-world setting, driving in the morning & afternoon same day	ambient light, normal road conditions
[35]	15 (9M, 6F), 20 - 67 years		The camera used in this study is an RGB camera and its frame rate is 10 frames per second Its commercial name is Raspberry Pi	BiosignalPlux Explorer chest belt	The study was conducted in the real world in a car	movement caused by driving activities or talking, changes in lighting, and different physical attributes of the test subjects such as height and weight
[4]	19 (13M, 6F)20-67 years		The cameras used in this study are the red, green, and blue (RGB) camera from Raspberry Pi Foundation in Cambridge, United Kingdom They have a frame rate of 10 FPS and are connected to a single-board computer Their commercial name is not specified	The ECG sensor BiosignalPlux Explorer is used as the ground truth for physiological signals monitoring	real world(in Braunschweig, Germany), in a car	movement caused by the car and driver, changes in illumination, skin color, hair and beard cuts. The specific conditions tested for include city, highway, and countryside scenarios
[33]	9 (5M, 4F), 37 ±8 years		The cameras used in this study are the Logitech HD Pro Webcam C920 and they have a frame rate of 30 FPS	Firstbeat	real-world, participants used their own vehicles for their daily driving activities	lighting, movement, and night

generalizability. Additionally, the relatively low frame rate of 10 FPS for RGB video data may constrain the temporal resolution for detecting rapid physiological or behavioral changes during driving.

The In-Vehicle Vital Signs Monitoring Dataset (2023) [35] aims to improve health assessments using a sensor system (accelerometer, piezoelectric sensor, camera) to monitor vital signs under various driving conditions. It provides data for algorithm improvement and result reproduction. This dataset includes 15 subjects (11 males and 4 females), with data recorded using an RGB camera at 10 FPS. Reference

accelerometer and piezoelectric sensor signals were sampled at 200 Hz, collected in real-world driving conditions across various scenarios.

- **Focus:** This dataset collects data from a sensor system designed to monitor vital signs in various driving conditions, focusing on improving health assessments from non-intrusive sensors.
- **Pros:** Helps in the enhancement of algorithms that must work with less conventional data types like accelerometer and piezoelectric sensor outputs, broadening the scope for non-invasive health monitoring in vehicles.

TABLE 7. Comparison between Machine Learning (ML) algorithms in included studies that is extracting rPPG, including computational efficiency.

Ref	ML	ROI	Features	Evaluation	Computational Efficiency
[34]	Neural Network	From chest and up	The acceleration, coordinates, Unix timestamp, beginning of the driving trip, user ID, and vital signs collected using a smartwatch	MSE, impact of illumination, driver movement, and car vibrations on accuracy	Moderate, requires lightweight hardware
[38]	Multitask cascaded convolutional network (MTCNN)	Face	Facial features and frequency spectrum features	MSE, RMSE, success rate of I (SUCI), and motion error rate (MER)	High due to cascaded architecture
[37]	Vision Transformer	Face	Encoded video frames	MAE, RMSE, Pearson's R	High computational cost (requires GPU acceleration)
[36]	CNN	Face	Spatio-temporal features from the fusion of RGB and NIR images	MAE, RMSE, SUC10, SUC20	Moderate, optimized for real-time processing
[32]	Transformer with Multi-Skip Connection Decoder	Face	Spatial and global features	MAE, RMSE	High computational cost due to transformer architecture
[31]	Tree-Based Probabilistic Fusion Network	Between the driver's eyes and mouth	Frequency features from Chrominance signal	MAE, RMSE	Low, suitable for edge devices
[39]	AdaBoost algorithm with improved CNN	Rectangular area of the forehead	Local features from the face, non-linear expressions	Bland-Altman analysis	Moderate
[35]	1D Convolutional Neural Network	Chest	Respiration signals from piezoelectric sensors, accelerometers, and a camera (rPPG, fusion)	positive predictive value (PPV), Sensitivity	Moderate, efficient for embedded systems
[4]	CNN-based Information Fusion Algorithm	Cheeks	R-wave in ECG and systolic peak in PPG	Mean performance percentage for ECG, PPG, iPPG sensors	High, requires GPU for real-time fusion
[33]	Random Forest, Deep Forest, and Neural Networks	Face	Action Units (AU), frame-wise activations, and time features	Balanced accuracy	Low, efficient for edge deployment

- **Cons:** The dataset includes a small and unbalanced sample size, with limited female representation, which could affect the generalizability of the findings. Additionally, the RGB camera's frame rate of 10 FPS and the 200 Hz sampling rate for accelerometer and piezoelectric sensors might limit the resolution for capturing rapid physiological or behavioral changes during driving scenarios.

Each dataset provides a unique set of features and data types that cater to specific research needs in the domain of driver monitoring and health assessment in automotive settings.

Datasets that include video footage of the driver and gold standard for physiological signals, such as MERL-rice NIR pulse (driving) [41], enable the extraction and analysis of rPPG signals and are thus featured in Table 8. In contrast, datasets lacking any key component for rPPG analysis were not included in Table 8, such as the 3MDAD dataset [42] and the UPCT dataset [13]. The 3MDAD dataset [42] was not included due to the absence of physiological signal measurements, which are necessary for gold standard validation in rPPG studies. In a similar vein, the UPCT dataset [13], despite its sophistication, it does not facilitate the direct application of rPPG methods due to the absence of facial video data, as the deployed RGB-D camera is oriented externally to capture street views and routes, not the driver's face.

Reflecting on Table 8, it's evident that female subjects are notably underrepresented in publicly available datasets for rPPG signal extraction from car drivers. Specifically, among the five public datasets examined, four had a scarcity of female participants. In total, only 14 female subjects were included in experiments conducted in real-world car settings, compared to 61 male subjects.

C. APPLICATIONS OF rPPG IN DRIVER STATE MONITORING

This subsection explores the application of remote photoplethysmography (rPPG) in driver state monitoring, highlighting its use in detecting fatigue, drowsiness, and drink driving, analyzing emotional states, and assessing distraction and mental load. These studies showcase rPPG's potential to enhance road safety by providing non-intrusive, real-time insights into driver well-being and cognitive states.

1) DETECTING DRIVER FATIGUE, DROWSINESS AND DRINK DRIVING

This section reviews several pivotal studies employing machine learning techniques for the detection of driver fatigue, drowsiness, and intoxication. Table 9 provides a detailed summary of these machine learning methods, showcasing their application across different datasets, their specific focus (e.g., fatigue, drowsiness, or drunk driving), and the outcomes achieved in terms of detection accuracy and other relevant metrics. Additionally, the diverse datasets used in these studies are outlined in Table 10, which details the source, demographic information of participants, and environmental conditions of each dataset. Together, these tables offer a comprehensive overview of the current landscape in detecting fatigue, drowsiness, and drink driving research, illustrating the breadth of data and methodological approaches currently in use.

Chen et al. [22] work presents a novel system designed to monitor and detect drink driving in real-time without interfering with the driving process. Utilizing a combination of vital signs extracted from WiFi signals and psychomotor coordination data obtained through steering wheel operations,

TABLE 8. Overview of publicly available datasets for noncontact physiological signal measurement.

Dataset	Year	Feature	Subjects	Resolution	Gold Standards	Environment
A multimodal dataset for various forms of distracted driving [40]	2017	Multimodal dataset acquired in a driving simulator under various distraction conditions. Included variables are speed, acceleration, brake force, steering, lane position, perinatal EDA, palm EDA, heart rate, breathing rate, facial expressions, and eye tracking data.	68 (35M, 33F)	FireWire CCD monochrome zoom camera: 1024×768 pixels, 15FPS. Thermal camera Tau 640 camera 640×512 pixels, 7.5 FPS	Physiological signals (heart rate, breathing rate, EDA) Shimmer3 GSR sensor for EDA, Zephyr BioHarness 3.0 for heart rate and breathing rate	Controlled(Lab), Simulator
MERL-rice NIR pulse (driving) [41]	2020	The first publicly available video dataset featuring ground truth pulse signals collected during driving. Driving with different motion Different filters for ambient light: 185 video	18 (16M, 2F)	RGB Camera: 640X640pixels 30FPS. NIR Camera: 640X640pixels 60FPS	A CMS50D+ finger pulse oximeter was used to capture the ground-truth PPG waveform, recorded at a rate of 60 frames per second (fps).	Real-World, Car
DriverMVT [12]	2022	Dataset for real-time monitoring of drivers, with videos collected in diverse driving scenarios including metadata like geographic coordinates, speed, and light conditions. Using USB and smartphone cameras under varied conditions.	9 (7M, 2F)	USB Camera: 30FPS, Smartphones: 1080x1920, 60 FPS	Heart rate (Xiaomi Mi Band 3)	Real-World, Car
In-Vehicle Health Monitoring Dataset [4]	2023	Dataset with ECG, PPG, and RGB video for continuous health monitoring in various driving scenarios. Includes face recognition for real-time face landmarks detection and medical analysis.	19 (13M, 6F)	RGB camera: 10 FPS	The sampling rate for the reference ECG, PPG, and Steering wheel ECG is 500 Hz	Real-World, Car (Volkswagen Tiguan)
In-Vehicle Vital Signs Monitoring Dataset [35]	2023	Developed to improve health assessments using a sensor system (accelerometer, piezoelectric sensor, camera) for monitoring vital signs in various driving conditions. Provides data for algorithm improvement and result reproduction.	15 (11M, 4F)	RGB camera: 10 FPS	The sampling rate for the reference, accelerometer, and piezoelectric sensor is 200 Hz	Real-World, Various Driving Conditions

the system employs a self-attention convolutional neural network, specifically the C-Attention model, to analyze these inputs. Conducted in a controlled laboratory setting with 15 participants, the study demonstrates a high accuracy in detecting drink driving (96.6%) and predicting blood alcohol content (BAC) with a mean absolute error between 0.002% and 0.005%. This contactless, non-invasive approach underscores the potential for integrating such technology into vehicles to enhance road safety by preventing drink-driving incidents.

Naurois et al. [23] work explores the utilization of Artificial Neural Networks (ANNs) to both detect and predict driver drowsiness in a simulated driving environment. Conducted with 21 participants using a highly controlled driving simulator, the study meticulously recorded physiological, behavioral, and driving performance data. Two ANNs were developed: one to determine the level of drowsiness at any given minute and another to predict the onset of drowsiness to a specified level. The findings indicate that integrating behavioral indicators with participant information such as driving time significantly enhances the prediction accuracy. The study achieved a detection mean squared error (MSE) of 0.22 and a prediction MSE of 4.18 minutes for reaching a moderate drowsiness level. This research contributes to the automotive safety field by providing a method that leverages non-intrusive data collection techniques for real-time monitoring and prediction of driver drowsiness, suggesting a potential for improved in-vehicle drowsiness detection systems.

Ramani et al. [24] developed a real-time drowsiness detection system designed to enhance road safety by preventing accidents caused by driver fatigue. Utilizing a low-power AI microcontroller and an AI camera, the system employs Convolutional Neural Networks (CNN) to analyze visual indicators of fatigue such as yawning and eye closure. The system was trained on 1310 images and tested in a live setting with ten participants. The research demonstrates the potential of integrating advanced AI technologies in real-world applications, providing a cost-effective solution for improving driver safety. This work contributes to the ongoing efforts in cognitive computing and cyber-physical systems, particularly in the automotive industry where safety and non-intrusiveness are paramount.

In their study, Kong et al. [25] introduce a novel approach to driver fatigue detection through the integration of remote photoplethysmography (rPPG) and motion tracking with a convolutional neural network and bidirectional long short-term memory (BiLSTM) framework. This research, detailed in their paper “Remote Photoplethysmography and Motion Tracking Convolutional Neural Network with Bidirectional Long Short-Term Memory: Non-Invasive Fatigue Detection Method Based on Multi-Modal Fusion,” proposes a multi-modal fusion method aimed at enhancing the accuracy and robustness of fatigue detection systems. Utilizing a self-created Multi-Modal Driver Alertness Dataset (MDAD), the study achieves an impressive accuracy rate of 98.2%. The system is tested under varied lighting and motion conditions to ensure its efficacy in real-world scenarios, marking

TABLE 9. Summary of machine learning methods used for detecting driver fatigue, drowsiness and drink driving.

Ref	Detection	Dataset (Public/Custom)	ML	ROI	Features	Evaluation	Accuracy
[22]	drunk driver	custom (self-made)	C-Attention CNN	chest	Breath rate, heartbeat rate, psychomotor coordination (angular velocity and acceleration from steering operations)	Accuracy, MAE, F1-score	96.60%
[23]	driver drowsiness	custom (self-made)	ANN	eyelid and head	heart rate and its variability, blink duration and frequency, PERCLOS, head movement in translation and rotation, steering wheel angle, lateral distance from lane	mean square error(MSE), absolute error thresholds, Pearson's correlation	Detection: 0.22 MSE
[24]	driver drowsiness	custom (self-made), Kaggle: Yawn-eye-dataset-new	CNN	eyes, mouth	Eye aspect ratio, blinking, yawning	Accuracy	80%
[25]	driver fatigue	custom (self-made : MDAD)	1D-CNN, BiLSTM ,	cheek, forehead	ECG HR, HRV, rPPG HRV, PERCLOS, yawn frequency, and a combination of PERCLOS, yawn frequency, HRV, and rPPG.	sensitivity, true positive rate , and accuracy	98.20%
[26]	driver fatigue	custom (self-made), public drowsy driving dataset	RNN	face, eyes	heart rate, eye openness level, and mouth openness level.	accuracy	94.70%
[27]	driver fatigue	NTHU-DDD	CNN	eyes, mouth and head	eye blinking, mouth yawning, and head movement.	accuracy	93.30%
[28]	driver fatigue	NTHU-DDD, UTA-RLDD, and YawnDD	CNNs and LSTM	eyes, mouth, and head	Visual features(eye features, mouth features, head features), physiological features (HR and BR)	Accuracy and loss	88.97%
[29]	driver fatigue	UTA-RLDD	SVM, FCNN, LSTM and the 1D CNN	face, eyes	PERCLOS, heart rate	accuracy, Precision	86.40%
[30]	driver fatigue	custom (self-made)	InceptionV3, LSTM	face	heart rate variability and time-series features of the eyes.	Precision, recall rate, F1 score	96.50%

a significant advancement in non-invasive technology for traffic safety and smart transportation systems. This method's integration of rPPG and motion tracking with advanced machine learning techniques represents a critical step forward in developing more reliable and effective driver assistance technologies.

Du et al. [26] introduced a novel non-intrusive method for detecting driver fatigue by integrating physiological and facial expression data using an RGB-D camera. This method utilizes a multimodal fusion recurrent neural network (MFRNN) to enhance detection accuracy through temporal analysis and fuzzy reasoning, which effectively handles noise in physiological signals. Tested under both simulated and real-world driving conditions, the system demonstrated superior performance compared to traditional methods. Its ability to operate non-intrusively and resist environmental variables such as light and obstructions represents a significant advancement in fatigue detection technology. The results indicate that this fusion method excels in noise reduction and fatigue prediction. Notably, the study involved sleep deprivation to induce fatigue and was tested with just one subject in a simulated driving environment.

Du et al. [29] developed a non-intrusive method for detecting driver fatigue using a multimodal fusion of eyelid movement and heart rate signals from RGB video footage. This approach employs a one-dimensional Convolutional Neural Network (1D CNN) to analyze PERCLOS (percentage of eyelid closure over the pupil) and heart rate data, utilizing the UTA-RLDD dataset, which includes RGB videos labeled according to the Karolinska Sleepiness Scale. By integrating outputs from both PERCLOS and heart rate models, the study aims to enhance fatigue detection accuracy. Despite the dataset not being captured in a car-like

environment as claimed, which could affect the findings' real-world applicability, the diverse demographic profile of the participants—varying in ethnicity, age, and physical characteristics like facial hair and glasses—strengthens the models' robustness and applicability. This diversity is crucial for ensuring the system performs reliably across different individuals and conditions.

Kassem et al. [27] developed a framework for predicting driver fatigue using non-contact sensors and Convolutional Neural Networks (CNN), achieving a 93.3% accuracy in detecting early signs of fatigue through facial and head behavior analysis. However, the study's limitations are notable: the experiments were conducted in simulated driving scenarios that do not fully replicate real-world conditions, potentially compromising the accuracy of facial landmark detection in uncontrolled environments. Additionally, the dataset included acted expressions and behaviors meant to simulate fatigue, which may not accurately represent genuine fatigue signs, thus affecting the framework's real-world applicability. These issues highlight the need for further refinement to ensure the framework's robustness and effectiveness in naturalistic settings.

Li et al. [28] introduce a novel fatigue detection method that combines physiological and visual data using advanced neural networks, including CNNs and LSTMs. This multimodal approach leverages the strengths of wearable devices and facial recognition technologies to accurately detect fatigue in real-time under various driving conditions. The method, showing a fatigue detection accuracy of 88.97% with forward-looking video compared to 83.01% with side-looking video, demonstrates high potential for improving road safety. However, the reliance on specific hardware and the variable quality of data due to environmental factors

pose challenges, particularly in maintaining the system in real-world automotive settings. Despite these limitations, this research significantly advances fatigue management systems and offers a foundation for further refinement.

Du et al. [29] demonstrated that their multimodal feature fusion approach, which integrates PERCLOS and heart rate features, outperforms traditional methods in detecting driver fatigue. Despite the study's claim of using a car-like environment, the dataset was not captured under such conditions. Nevertheless, the diverse composition of the dataset, which includes subjects of various ethnicities, ages, and physical characteristics such as facial hair and glasses, adds to the robustness and applicability of the model. This variety ensures that the fatigue detection system is tested across a broad spectrum of visual and physiological cues, enhancing its reliability in real-world scenarios.

Zhao et al. [30] developed a sophisticated method for detecting fatigue and sleepiness, combining the InceptionV3 convolutional neural network with Long Short-Term Memory (LSTM) networks. This approach integrates traditional image processing with heart rate variability (HRV) data extracted from video footage using remote photoplethysmography (rPPG), effectively identifying signs of fatigue. The study's dataset was strategically collected at three different times of the day—7:00 AM, 1:00 PM, and 10:00 PM—to capture the natural fluctuations in human alertness. This timing ensures the model's effectiveness under conditions where fatigue is likely, enhancing its real-world applicability in transportation safety and workplace monitoring. The integration of image-based and physiological signals through advanced machine learning algorithms demonstrates a clear link between heart rate and fatigue, showcasing significant advancements in non-invasive fatigue detection technology.

In conclusion, the studies reviewed consistently demonstrate that multimodal fusion significantly enhances the accuracy of detecting driver fatigue, drowsiness, and drink driving. This approach, integrating various physiological and behavioral data, has been tested under multiple conditions and across diverse subject ethnicities, affirming its robustness. However, it is noteworthy that most of these studies have been conducted in simulated environments. While this allows for controlled experimentation, it may limit the applicability of findings to real-world driving scenarios. Future research should aim to bridge this gap by validating these technologies in naturalistic settings to confirm their effectiveness and reliability on the road.

2) DRIVER EMOTIONAL AND MENTAL STATE MONITORING

This section examines the applications of remote photoplethysmography (rPPG) in monitoring both the emotional and cognitive states of drivers, crucial for ensuring optimal driving performance and safety. The studies addressed include advancements in recognizing diverse emotional responses ([14], [15], [16]), and assessments of mental states such as driver distractions, mental workload, and cognitive

load ([17], [18], [19], [20]). Additionally, the impact of these states on takeover performance in autonomous driving scenarios is explored ([21]). Collectively, these studies underscore the potential of rPPG technology in providing real-time, non-intrusive monitoring that can significantly contribute to the safety and efficiency of modern driving environments.

Tables 11 and 12 provide a detailed summary of the methodologies and datasets used in these studies. Table 11 focuses on the machine learning methods employed for detecting emotional and mental states, highlighting the detection type, dataset, machine learning algorithms, region of interest (ROI), features extracted, evaluation metrics, and accuracy. Table 12 offers an overview of the datasets used, detailing the number and demographic information of subjects, camera specifications, ground truth methods, experimental environment, and data conditions. These tables collectively offer a comprehensive view of the current state of research in driver state monitoring using rPPG and related technologies.

Rahman et al. conducted two studies in this domain. In the initial study [20], they explored the classification of cognitive load in vehicular drivers using non-contact methods, concentrating on physiological and vehicular parameters. The research utilized high-resolution RGB cameras and vehicle data, eschewing traditional wire-based sensors to minimize driver disruption. It involved 63 male participants aged 35 to 50 years, with data collected in a driving simulator. Four machine learning algorithms were employed: logistic regression (LR), support vector machine (SVM), linear discriminant analysis (LDA), and neural networks (NN). Physiological parameters were extracted from facial videos, while vehicular parameters were acquired through the controller area networks (CAN). The study concluded that the LR model achieved the highest accuracy, recording 94% in the first study and 82% in the subsequent one. The camera data yielded higher accuracy than traditional sensors due to fewer artifacts, demonstrating the efficacy of non-contact methods in dynamic driving environments.

In the subsequent study [19], Rahman et al. investigated driver cognitive load classification using a blend of machine learning and deep learning techniques. Thirty-three male participants, aged 35 to 50 (average 42.47 ± 4.39), participated in this research conducted in a car simulator (VTI Driving Simulator III). Eye movements were recorded using both a high-resolution RGB camera operating at 30 fps and the SmartEye eye-tracking system, which served as the ground truth. The driving session lasted 40 minutes, including a 10-minute practice session, with participants engaged in one-back cognitive load tasks during the driving scenarios. Each scenario was repeated four times, gathering data during baseline and cognitive load tasks. A custom dataset generated for this study trained and evaluated several machine learning models (SVM, LR, LDA, k-NN, DT) and deep learning architectures (CNN, LSTM, AE). The focal point was the eye

TABLE 10. Overview of datasets used in fatigue, drowsiness and drunk driver detection studies.

Dataset	Subjects	Age (years)	Ethnicity	Camera	Ground truth	Environment	Data/conditions
UTA-RLDD [43]	60(51M, 9F)	20 to 59	10 White, 5 non-white Hispanic, 30 IndoAryan and Dravidian, 8 Middle Eastern, and 7 East Asian	participants' mobile phones or web cameras.	participants' self-assessed state of alertness or fatigue using Karolinska Sleepiness Scale (KSS)	Videos were taken from roughly different angles in different real-life environments and backgrounds.	30 hours, 180 RGB videos, three different drowsiness states
NTHU-DDD [44]	36 (18M, 18F)	18 to 40	Diverse, including various skin colors described as black or brown, white, and yellow	D-Link DCS-932L for IR videos; Logitech C310 HD for RGB videos, 30 frames per second.	acted expressions and behaviors to represent drowsiness and non-drowsiness	Daytime and nighttime settings to simulate various driving conditions.	Videos captured under various scenarios including with and without glasses or sunglasses, different lighting conditions
YawnDD [45]	107 (57M, 50F)	16 and above	White, African, Middle-eastern, and Asian.	video 640x480 24-bit true color (RGB) 30 fps AVI without audio	acted normal driving, driving while talking/singing, and driving while yawning	in the car during day time (various lighting situations), sunny to rainy	set1 (322 videos), set 2(29 videos), without glasses, men with and without beard, men with and without moustache, women with and without scarf, different hairstyles and different clothing
[30]	46 gender not mentioned	not mentioned	Not specified but the images show Asian	frame rate of 30 fps and a resolution of 1980 ×1020 pixels.	ECG	Driving simulator: Euro Truck Simulator 2 game	simulated driving situations in the game Euro Truck Simulator 2, at three different times during the day (7:00 AM, 13:00 PM, and 22:00 PM). The subjects wore long-range electrocardiograms and their faces were recorded for a full hour. The videos were recorded in both indoor backgrounds and vehicles.
MDAD (Multi-Modal Driver Alertness Dataset) [25]	52(28M, 24F)	22 to 53	Not specified but the images show Asian	Oni S500 infrared camera (simultaneously captures RGB and infrared images at a sampling rate of 30 Hz, resolution of 640 × 480)	ECG for Fifteen participants (7 in the morning and 8 in the evening)	real world driving: scenarios involving different types of vehicles (private cars, taxis, trucks, etc.). Daytime and nighttime lighting conditions are tested.	4000 (1200 × 650 RGB-IR) driving videos. Among these, 2530 are labeled as awake, while 1470 are labeled as fatigued
[26]	23(14M, 6F)	20 and 34	Not specified but the images show Asian	RGB-D camera was employed to capture both RGB and infrared videos, 1920 × 1080 pixels.	for Fatigue: Karolinska sleepiness scale (KSS), EEG, self-assessment. For Heart Rate: Polar H10.	Driving simulator: game, real world driving: one participant	effects of sleep deprivation (nonsleep-deprived vs. sleep-deprived groups), ambient light conditions, and glasses usage
public drowsy driving dataset [46]	23 (12M, 11F)	26 to 55	Not specified	300 fps camera	Stanford Sleepiness Scale, Epworth Sleepiness Scale, hours of sleep	Driving simulator	Videos during reading, watching driving simulation, and performing driving simulation; less than 30 min per participant
[22]	15 (10M, 5F)	21 to 26	Not specified	None	Neulog236 belt, ECG, Breathalyzer	Driving simulator	Drink effects on vital signs and psychomotor coordination
[23]	21 (11M, 10F)	24.09 ± 3.41	Not specified	Webcam	Trained Observer Rating	Driving simulator	Driving simulator sessions of 110 min, behavioral and physiological data collection, induced drowsiness conditions
[24]	10	Unspecified	Not specified	Webcam	NA	NA	variables like camera distance and illumination effects

pupil, with statistical features extracted from saccades and fixations. The highest accuracy achieved was 92% using the SVM model.

In the context of these two studies by Rahman et al. [19], [20], the 1-back and 2-back tasks are cognitive load exercises employed to evaluate working memory and attention. In the 1-back task, participants are presented with a sequence of stimuli—such as numbers, letters, or images—and must indicate when the current stimulus matches the one immediately preceding it. For instance, if the sequence is A, B, C, B, participants should respond when the second B appears, as it matches the previous one. The 2-back task, more

challenging, requires participants to identify when the current stimulus matches the one that appeared two steps earlier in the sequence. For example, if the sequence is A, B, A, C, B, participants should respond when the second A appears, as it corresponds to the one two steps before. These tasks are widely utilized in cognitive research to impose varying levels of mental workload and to assess participants' working memory capacity and cognitive flexibility.

M. Gjoreski et al. [17] investigates the use of machine learning (ML) and deep learning (DL) algorithms to detect driver distractions using physiological and visual signals. Data were collected from 68 male volunteers in a driving

TABLE 11. Summary of machine learning methods in driver emotional and mental state monitoring studies.

Ref.	Detection	Dataset (Public/Custom)	ML	ROI	Features	Evaluation	Accuracy
[20]	Mental	custom (self-made)	LR, SVM, LDA, NN	Face	HRV features (time, frequency, and non-linear domains) and vehicular features (speed, steering, yaw rate, torque, lane position, reversal rate, lane exit, mean squared error).	sensitivity, specificity, accuracy, precision, and recall	94%
[19]	Mental	custom (self-made)	SVM, LR, LDA, k-NN, DT, CNN, LSTM, and AE	Eye pupil	statistical features for saccades and fixations derived from eye movement data.	Sensitivity, specificity, precision, F1-score, accuracy, and ROC AUC	92%
[17]	Mental	Public: A multi-modal dataset for various forms of distracted driving [40]	DT, RF, NB, KNN, SVM, AdaBoost, XGB, STRNet, CNN, LSTM	Face	statistical metrics from physiological signals (e.g., heart rate, breathing rate, electrodermal activity) and visual data (e.g., facial action units, emotional activations)	Precision, recall, F1-score, and average precision (AP)	F1-score: 94%
[18]	Mental	custom (self-made)	DT, DA, LR, SVM, kNN, Ensemble	Nose tip, Glabella	Statistical, frequency domain, and entropy measures from ECG and thermal IR.	Accuracy, sensitivity, specificity, precision, F1-score, AUC	75%
[21]	Mental	custom (self-made)	SVM, XGBoost, CNN, LSTM, MTCNN, MTTC-CAN	Head, face, eyes, body	Facial expressions, gaze direction, head orientation, and body posture measures from video data.	Accuracy, weighted F1-score	95%
[14]	Emotional	custom (self-made)	CNN, Bi-LSTM, MFB.	Face	Facial geometry, facial skin brightness changes for heart rate, and fused spatial-temporal features.	Accuracy	91.51%
[16]	Emotional	Citiescapes, Wild-Dash 2, custom (self-made)	DT, GBDT, XGBoost, LightGBM, CatBoost	Front view captured by a camera next to the driver's eyes	HRV-based indicators such as the mean value and standard deviation of NN intervals and environmental indicators such as the proportion and variation rate of 14 elements (e.g., trucks, buses, mountains, sky, cars).	F1-score and G-mean	F1 score: 85.5%, G-mean: 89.0%
[15]	Emotional	custom (self-made)	SVM, NN, RF, XGBoost, AdaBoost, GradBoost	cheeks, forehead	heart rate variability metrics including BPM, RMSSD, SDNN, and BSV from PPG signals.	precision, recall, accuracy, and macro-F1	97.80%

simulation, capturing signals such as heart rate, breathing rate, electrodermal activity, and facial expressions. Various statistical features were extracted from these signals, and both classical ML methods (e.g., decision trees, random forests, SVM, XGB) and DL models (e.g., CNN, LSTM, STRNet) were employed to classify driving sessions as distracted or not. The evaluation metrics included precision, recall, F1-score, and average precision, with the highest session classification F1-score reaching 94% using the XGB algorithm with 60-second windows of facial action units (AUs) as input. The study highlights the effectiveness of combining physiological and visual data to enhance driver distraction detection systems.

Cardone et al. [18] assess drivers' mental workload using a combination of physiological signals and machine learning techniques. Conducted with 26 participants driving in a simulated urban environment, the research collected ECG signals and thermal infrared (IR) videos to monitor physiological responses. Participants' mental workload was manipulated and measured using two cognitive tests: the Digit Span Test (DST) and the Rey Auditory Verbal Learning Test (RAVLT). The DST assessed short-term and working memory through forward and backward repetition of digit sequences, while the RAVLT evaluated verbal learning and long-term memory through immediate and delayed recall and recognition of word lists. Various machine

learning algorithms, including Decision Trees, Discriminant Analysis, Logistic Regression, Support Vector Machines, Nearest Neighbor, and Ensemble Classifiers, were employed to analyze the data. Features extracted for classification included statistical measures, frequency domain features, and entropy measures from both ECG and thermal IR data. The highest accuracy achieved was 75.0% with a multimodal approach combining IR and HRV features. This study highlights the feasibility of using non-invasive techniques to classify drivers' mental workload accurately, providing valuable insights for the development of advanced driver assistance systems (ADAS) aimed at improving road safety.

AffecTake is an innovative emotion-aware takeover performance prediction system designed specifically for semi-autonomous driving environments [21]. It utilizes a three-stage framework—sensing, predicting, and responding—to enhance the interaction between the vehicle and the driver. During the sensing stage, an in-vehicle camera captures comprehensive multi-channel data that includes the driver's facial expressions, head pose, eye movement, body posture, and heart rate, using advanced deep learning technologies. This data provides insights into both the emotional and physical state of the driver, facilitating a nuanced understanding without the need for invasive wearable devices.

TABLE 12. Overview of (custom) Datasets used in driver emotional and mental state monitoring studies.

Dataset	Subjects	Age (years)	Camera	Ground truth	Environment	Data/conditions
[20]	63(63M,0F)	35 to 50 (42.47 ± 4.39)	30 fps, RGB, high resolution	g.tec sensor (ECG) & DeweSoft software	car simulator (VTI Driving Simulator III)	40-min sessions (10-min practice), 1-back and 2-back tasks scenarios repeated 4 times, data from baseline and cognitive load tasks.
[19]	33(33M,0F)	35 to 50 (42.47 ± 4.39)	30 fps, RGB, high resolution	SmartEye eye-tracking system, which tracked and captured the eye movements of the drivers.	car simulator (VTI Driving Simulator III)	40-min sessions (10-min practice), 1-back task scenarios repeated 4 times, data from baseline and cognitive load tasks.
[18]	26 (17M, 9F)	18 to 42 (30.89 ± 6.08)	Intel RealSense D415 depth camera, FLIR Boson 320LW IR thermal camera	ECG lead Encephalan Mini (Medicom MTD sys) ,256HZ	Driving simulator: City Car Driving, Home Edition	45 min urban context, adverse conditions, participants' behavioral and physiological responses were monitored using Digit Span Test (DST) and Rey Auditory Verbal Learning Test (RAVLT).
[21]	26 (14M, 12F)	20 to 43	Rapoo C260AF webcam, 720p resolution, RGB, 30 fps.	"Self-Assessment Manikin (SAM) questionnaire & experimental observer annotations	Driving simulator with CARLA software	70 minutes, four driving courses per participant, each containing three takeover events, resulting in a total of 312 driving videos, behavioral and physiological data including facial expressions, heart rate, body posture, eye movement, and head pose.
[14]	12(9M,3F)	NA	Kinect v2, RGB, resolution 1920x1080, 30 fps	BMD 101 ECG Sensor, 256Hz	Driving simulator with the Kinect placed right in front of the driver to capture the face of the driver.	videos from 12 volunteers, each video was cut into a 1-minute clip
[16]	21 (21M, 0F)	(33.05± 6.20)	camera is a module of an eye tracker	12-lead ECG recorder	Real-world driving on a high-altitude expressway, clear days, 70 km section, 4,000 meters altitude, varied lighting and visibility.	3712 sets of driving environment data, 56 indicators, stress levels: low, medium, high.
[15]	21	20 to 54	PiCamera, 2592 ×1944 in YUV colour space at 15 fps	PPG contact sensor, GSR sensor	Simulation to mobility solution Gyrolift: wheelchair connected to a screen with force feedback generating vibrations.	2221 samples from 21 participants watching a 6-minute Stroop test video; each sample includes four parameters (BPM, RMSSD, SDNN, BSV) from PPG signals indicating stressed or relaxed states.

Takeover readiness, a key concept in this system, refers to the driver's preparedness to assume control of the vehicle when a takeover request (TOR) is issued. It is crucial for the transition from automated to manual driving, ensuring the driver is both physically and mentally prepared to handle the vehicle. AffecTake processes the captured data to predict the driver's readiness and reaction time for taking over control. Using specialized deep learning models, AffecTake achieves high accuracy in predicting takeover performance, demonstrating a 95% accuracy and a 0.94 weighted F1 score for takeover readiness, along with a 91.67% accuracy and a 0.9 weighted F1 score for takeover time [21]. The responding stage of the system uses these predictions to initiate safety measures such as warnings or other interventions, ensuring that the driver is adequately prepared to take over control. This holistic approach not only bolsters safety but also enhances the efficiency of semi-autonomous vehicles by ensuring smooth transitions between automated and manual control.

In a study by Du et al. [14], they proposed a novel deep learning framework called Convolution Bidirectional Long Short-Term Memory Neural Network (CBLNN) for

real-time driver emotion recognition. The method combines facial geometry and heart rate features extracted from facial skin brightness changes using Kinect v2. The study involved 12 volunteers in a simulated driving environment, capturing 1-minute video clips to analyze emotions. It is not explicitly detailed in the study how emotions were induced in the simulated driving environment. The CBLNN framework integrates CNN for facial feature extraction and Bi-LSTM for temporal heart rate analysis, with a final classification using Multi-modal Factorized Bilinear pooling (MFB). The evaluation metrics included accuracy, with the highest accuracy achieved being 91.51% for recognizing "sad" emotions. The study highlights the potential of combining spatial and temporal features for improving emotion recognition accuracy in driving scenarios.

Qin et al. [16] aimed to predict driving stress levels by analyzing environmental features and heart rate variability (HRV) indicators. Conducted on a 70 km section of the G6 Beijing–Lhasa Expressway, the study involved 21 male drivers. Environmental features were captured using a camera placed next to the driver's eyes, and HRV data were collected via ECG. The input features for the model included

various HRV metrics and proportions and variation rates of 14 environmental elements. Stress levels were classified into low, medium, and high using exploratory factor analysis and K-means clustering. The study employed machine learning algorithms such as XGBoost, which achieved the highest performance with a macro-average F1 score of 0.855 and a G-mean of 0.890. This approach demonstrated the feasibility of using environmental elements and HRV indicators to assess and predict driving stress in real-time, providing a less intrusive and safer method for driving stress evaluation.

Tagnithammou et al. [15] assessed the feasibility of using a low-cost camera to estimate the emotional state of individuals on a mobility solution, similar to the Gyrolift. The experiment involved 21 participants who watched a 6-minute Stroop test video designed to induce stress, during which Photoplethysmography (PPG) signals were recorded to extract heart rate variability metrics. Each collected sample includes four parameters: Beats Per Minute (BPM), Root Mean Square of Successive Differences (RMSSD), Standard Deviation of NN intervals (SDNN), and Balance of Sympatho-Vagal (BSV) from PPG signals. The machine learning algorithms used included Support Vector Machine (SVM), Neural Network (NN), Random Forest (RF), Extreme Gradient Boosting (XGBoost), Adaptive Boosting (AdaBoost), and Gradient Boosting (GradBoost), with the highest accuracy of 97.8% achieved using XGBoost. The study demonstrated that it is possible to use a camera to classify emotional states with significant accuracy, suggesting potential applications for improving user interaction and safety in mobility devices.

The review of literature on Driver Emotional and Mental State Monitoring reveals significant progress in leveraging remote photoplethysmography (rPPG) for enhancing driver safety and performance. The cited studies underscore the efficacy of rPPG technology in real-time, non-intrusive monitoring of both emotional and cognitive states, which are critical for maintaining optimal driving conditions. The integration of machine learning and deep learning techniques with physiological and visual data has proven effective in detecting a range of driver states, from emotional responses to cognitive load and distraction levels.

The detailed summaries provided in Tables 11 and 12 offer a comprehensive overview of the methodologies and datasets employed in these studies. The emphasis on machine learning methods and diverse datasets highlights the adaptability and potential of rPPG technology across various driving scenarios and conditions. Furthermore, the exploration of the impact of these mental states on takeover performance in autonomous driving scenarios demonstrates the critical role of rPPG in the evolving landscape of vehicle automation.

Overall, the findings from these studies indicate a promising direction for future research and practical applications. The continuous development and refinement of rPPG-based monitoring systems can contribute significantly to advanced driver assistance systems (ADAS) and enhance the safety and efficiency of both manual and autonomous driving environments.

V. DISCUSSION

A. INTERPRETATION OF KEY FINDINGS

Recent advancements in machine learning (ML) algorithms have significantly enhanced the performance of rPPG systems, particularly in driver monitoring applications. Several algorithms, including deep learning (DL) and traditional machine learning techniques, have been employed for interpreting rPPG data to estimate physiological metrics like heart rate (HR) and heart rate variability (HRV) in real-time. Convolutional neural networks (CNNs) and their variants, such as multitask cascaded CNNs (MTCNNs), have shown superior performance in extracting rPPG signals from video streams due to their ability to handle the variability in illumination and motion commonly encountered in vehicular environments. Other models, such as Vision Transformers, have also demonstrated high accuracy, with results often outperforming traditional models like Random Forests or Support Vector Machines (SVM) in specific tasks like HR estimation [36].

In terms of accuracy and reliability, DL-based methods generally achieve lower mean absolute errors (MAE) and root mean square errors (RMSE) compared to traditional machine learning models. However, balancing computational efficiency and performance remains a challenge, particularly for real-time applications. Transformer-based models have emerged as promising candidates due to their ability to efficiently process spatial and temporal data. These models have been shown to maintain a good balance between performance and computational load, especially when deployed on embedded systems in vehicles [37].

Optimizing ML algorithms for non-ideal conditions, such as varying lighting or driver movements, has been a central focus of recent research. Techniques like adaptive signal fusion, which integrate signals from multiple sensors (e.g., RGB, near-infrared, and thermal cameras), have improved the robustness of rPPG algorithms against these challenges. Furthermore, advancements in error-compensation frameworks, which reduce noise in data caused by environmental changes, have yielded promising results in increasing rPPG's reliability under diverse driving conditions [38].

The most effective features extracted from rPPG data for driver monitoring include frequency spectrum features, temporal features such as HRV, and facial features (e.g., facial movements and skin color changes). These features, when combined with physiological signals like HR, provide a comprehensive representation of the driver's state, enabling more accurate fatigue and stress detection [4].

Machine learning-based rPPG systems have demonstrated high effectiveness in detecting driver fatigue, stress, and cognitive load. Compared to traditional physiological monitoring systems, which often rely on contact sensors like electrocardiograms (ECG), rPPG offers a non-contact alternative that does not disrupt the driver. While traditional methods like ECG and PPG still provide more accurate baseline data, rPPG systems augmented by machine learning techniques have been closing the accuracy gap, particularly in detecting

subtle physiological changes that correlate with fatigue and cognitive load [25].

Recent advancements in ML, particularly the integration of multimodal data (e.g., combining HRV with facial expression data), have enhanced rPPG's predictive accuracy and real-time analysis capabilities. These innovations allow rPPG systems to detect signs of driver fatigue or emotional stress earlier than conventional methods. Moreover, the use of CNN-based fusion models has resulted in real-time analysis of driver states, which is critical for ensuring timely interventions to prevent accidents caused by fatigue or distraction [26].

B. THEORETICAL AND PRACTICAL IMPLICATIONS

The findings from this review contribute significantly to the existing theories and models in rPPG and driver monitoring. By synthesizing recent advancements in machine learning (ML) algorithms applied to rPPG, our work enhances the understanding of how non-contact physiological monitoring can be effectively integrated into driver state detection systems. Specifically, the use of deep learning models, such as convolutional neural networks (CNNs) and transformers, to improve the accuracy and robustness of rPPG signal extraction under varying environmental conditions contributes to the theoretical framework of intelligent driver monitoring. These findings support and expand current models that emphasize the importance of multi-modal data fusion in enhancing the reliability and precision of rPPG-based systems, suggesting that combining rPPG with other physiological and behavioral signals is crucial for comprehensive driver monitoring.

The practical implications of our findings are profound, particularly for the development of Advanced Driver Assistance Systems (ADAS) and other real-world applications. The review highlights how advanced ML techniques can optimize rPPG systems for real-time driver monitoring, making them more viable for integration into commercial vehicles. These improvements could lead to more accurate detection of driver fatigue, stress, and cognitive load, thereby enhancing the effectiveness of ADAS in preventing accidents. Furthermore, the development of robust rPPG systems that function reliably under diverse driving conditions could drive the widespread adoption of non-contact monitoring technologies in both private and commercial transportation sectors, ultimately contributing to safer driving environments.

Looking back at Table 8, it is clear that female subjects are significantly underrepresented in publicly available datasets for rPPG signal extraction from car drivers. Among the five public datasets analyzed, four exhibit a notable lack of female participants. Specifically, only 14 female subjects were included in studies conducted in real-world car environments, compared to 61 male subjects. To address this gender disparity, it is essential to thoroughly investigate the factors contributing to the limited female participation in these datasets and to develop strategies that encourage greater involvement. Alternatively, advancing the generation

of synthetic rPPG signals using techniques such as GANs could provide a practical solution.

C. ETHICAL AND SOCIETAL CONSIDERATIONS

The use of rPPG technology in driver monitoring raises several ethical considerations, particularly concerning privacy and consent. Since rPPG relies on video-based monitoring, there is a risk of infringing on drivers' privacy, as continuous surveillance could capture more than just physiological data. Ensuring that rPPG systems are designed with strong privacy protections, such as data anonymization and encryption, is essential. Moreover, obtaining informed consent from drivers before implementing such monitoring systems is crucial. This requires transparent communication about how the data will be used, who will have access to it, and the measures in place to protect their privacy.

The broader societal impacts of implementing rPPG technology in driver monitoring extend beyond individual privacy concerns. The widespread adoption of such systems could lead to changes in driving behavior, as drivers may become more aware of their physiological states and adjust their driving accordingly to avoid detection of fatigue or stress. Additionally, insurance companies might incorporate data from rPPG systems into their risk assessments, potentially influencing insurance premiums based on drivers' monitored physiological data. Furthermore, regulatory frameworks may need to be updated to address the ethical and legal implications of using such technology, ensuring that it is deployed in a way that balances safety with individual rights.

D. LIMITATIONS OF THE REVIEW

This review, while comprehensive, is not without its methodological limitations. One such limitation is the potential bias in study selection, as the inclusion criteria may have inadvertently excluded relevant studies that did not meet specific search terms or were published in less prominent journals. Additionally, the quality assessment of included studies was based on the information available in the publications, which may not fully reflect the rigor of the methodologies used. This could lead to an overestimation or underestimation of the effectiveness of certain ML algorithms or rPPG systems.

The scope of this review is also limited by the exclusion of certain types of studies and technologies. For instance, studies focusing on non-contact physiological monitoring methods other than rPPG, such as radar-based systems, were not included, which may have provided additional insights into alternative approaches for driver monitoring. Furthermore, the review primarily focused on ML algorithms, potentially overlooking advancements in other areas of rPPG technology, such as hardware innovations or alternative data processing techniques. These exclusions suggest that while the review offers valuable insights into the current state of rPPG systems for driver monitoring, it may not fully capture the breadth of research and technological developments in this rapidly evolving field.

VI. FUTURE DIRECTIONS

A. TECHNOLOGICAL ENHANCEMENTS

The future of remote Photoplethysmography (rPPG) in driver monitoring systems is closely tied to technological advancements that address current limitations. One of the primary areas for development is the improvement of sensor technologies. Current rPPG systems often struggle with variability in environmental conditions, such as changes in lighting or motion artifacts caused by head movements. Future research should focus on developing more robust sensors capable of operating effectively across a wide range of conditions, including low light environments and during rapid movements. The integration of multi-sensor approaches, combining RGB, infrared, and thermal imaging, could provide a more reliable and accurate assessment of physiological states in these challenging scenarios.

Another technological enhancement lies in the refinement of signal processing algorithms. The ongoing development of machine learning models, particularly deep learning techniques like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), offers promising avenues for improving the accuracy of rPPG signal extraction and analysis. These models need to be optimized for real-time applications, ensuring that they can process data efficiently without compromising accuracy, even in dynamic driving conditions.

Future research should explore the integration of advanced model compression techniques, such as knowledge distillation and innovative architectures like Group Multi-view Vision Transformers (GMViT) [47], to enable real-time driver monitoring systems on memory-limited devices. These methods have shown potential in reducing parameter size and computational demands while maintaining high performance, making them ideal for in-vehicle deployment.

Further advancements can be achieved by leveraging innovations in feature extraction, such as Adjustable Jacobi–Fourier Moments (AJFM), which offer refined control over spatial information distribution [48]. Incorporating such techniques could enhance the versatility and accuracy of machine learning models in detecting complex driver behaviors.

B. EXPANDING REAL-WORLD APPLICATIONS

While rPPG technology has shown significant potential in controlled environments, its application in real-world scenarios remains underexplored. Future research should prioritize the testing and validation of rPPG systems in a broader range of driving conditions, including urban, rural, and highway settings. This expansion is essential for understanding how these systems perform in diverse environments, which are characterized by varying speeds, lighting conditions, and driver behaviors.

Moreover, there is a need for studies that include a more diverse population of drivers. Much of the existing research has focused on narrow demographic groups, which limits the

generalizability of the findings. Future studies should ensure that participant pools are representative of different genders, ages, ethnicities, and health conditions. This diversity is crucial for developing rPPG systems that are effective across the entire driving population.

C. ETHICAL CONSIDERATIONS AND DATA PRIVACY

As rPPG systems become more integrated into vehicles, ethical considerations will become increasingly important. These systems collect sensitive physiological data, which raises concerns about privacy and data security. Future research should explore methods for ensuring that rPPG data is collected and stored in a manner that protects individual privacy. This might involve the use of encryption, anonymization techniques, and secure data storage solutions.

In addition to privacy concerns, there is a need for research into the ethical implications of continuous physiological monitoring. For example, how will the data be used by insurance companies or law enforcement agencies? What are the potential consequences for drivers who are constantly monitored for signs of fatigue or stress? Addressing these questions will be crucial for ensuring that rPPG systems are implemented in a way that respects individual rights and freedoms.

D. INTERDISCIPLINARY COLLABORATION AND STANDARDIZATION

The development of rPPG systems for driver monitoring is inherently interdisciplinary, requiring collaboration between experts in fields such as computer vision, automotive engineering, human factors, and artificial intelligence. Future research should encourage such collaborations, which are essential for overcoming the complex challenges associated with in-vehicle monitoring.

Furthermore, there is a pressing need for the standardization of methodologies and evaluation metrics in rPPG research. Currently, the lack of standardized protocols makes it difficult to compare results across studies or to validate findings in different contexts. Establishing common standards for data collection, algorithm testing, and performance evaluation will be critical for advancing the field and ensuring that rPPG systems can be reliably implemented in real-world vehicles.

E. FUTURE RESEARCH OPPORTUNITIES

Finally, future research should explore the integration of rPPG with other emerging technologies, such as autonomous driving systems and advanced driver-assistance systems (ADAS). By combining rPPG with other monitoring technologies, such as eye tracking or EEG, it may be possible to develop comprehensive systems that provide a more holistic view of driver state and behavior. These integrated systems could significantly enhance vehicle safety by providing early warnings of driver impairment or distraction, thereby reducing the risk of accidents.

Moreover, research should investigate the potential of using rPPG data for long-term health monitoring, both within and outside of the driving context. Understanding how physiological signals change over time in response to different stressors or conditions could lead to new applications in health and wellness, potentially transforming rPPG from a tool for driver safety into a broader platform for personal health management.

VII. CONCLUSION

In conclusion, this systematic review has underscored the critical role of camera-based systems, particularly remote photoplethysmography (rPPG), in advancing non-invasive monitoring of driver physiological states. These technologies, driven by machine learning algorithms and intelligent signal processing, have shown significant potential in enhancing the safety and well-being of drivers by providing real-time insights into their health and alertness levels.

Our analysis of current literature reveals that while there are numerous advancements in the field, significant challenges remain, particularly in terms of adapting these technologies to the dynamic and often unpredictable conditions of the driving environment. Issues such as variable lighting, vehicle motion, and diverse physiological responses among drivers call for more robust, adaptive machine learning models and more resilient sensor technologies.

Furthermore, this review has identified several gaps in current research, most notably the need for larger, more diverse datasets that reflect the full spectrum of driving scenarios and driver demographics. There is also a pressing need for standardization in the methods used to evaluate the performance of these systems, ensuring that they are reliable and effective across various conditions and populations.

Looking forward, it is clear that fostering interdisciplinary collaboration among experts in machine learning, computer vision, automotive engineering, and human factors is essential for addressing these challenges. The integration of these diverse fields can lead to the development of more sophisticated, context-aware systems capable of more accurately and effectively monitoring driver states.

In conclusion, while the path forward is fraught with challenges, the potential benefits of successfully integrating rPPG and similar technologies into automotive safety systems are immense. As this field evolves, it is anticipated that the next generation of driver assistance systems will not only improve road safety but also enhance the overall driving experience by ensuring the health and alertness of drivers, ultimately leading to fewer accidents and safer roads.

VIII. DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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